Trial Balance Review
and AI Augmentation

Implementation Specification

Ref: `TB-REVIEW-2026-v1.2`

SAP-OSS Financial Review Team

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Abstract

This specification defines the target-state implementation of Standard Chartered Bank's Trial Balance (TB) review process across all Legal Entities, Segment, and Group, together with the AI-augmented operating model sanctioned under the business case approved on 2025-12-14 (requirement `TB-REQ-001`, author S1 Badrinath). The document captures the process as it is currently executed by the Global Financial Services (GFS) Analyst and the Head of Africa GFS under three governing End User Computing (EUC) controls, and specifies the automation envelope that replaces the manual variance analysis, commentary generation, and FORTM pack assembly.

Three orthogonal workstreams are specified in full:

- a **canonical TB corpus** of workflow, control, decision, requirement, KPI, and glossary records, each validated against a JSON Schema in `structured/schema/`;
- an **AI augmentation pipeline** that ingests PSGL / S/4 extracts, performs statistical anomaly and materiality detection (percentile-based rules in v1.0; machine-learning methods scoped for v2.0, see Chapter 10), drafts variance commentaries with a vLLM-hosted model, and generates the monthly FORTM pack narrative;
- a **workflow and controls engine** that formalises the 28-state BPMN flow extracted from the GLO control narrative and binds the three EUC controls (`EX/200/1353107/0001/000`, `EX/200/1579702/0001/000`, `EX/200/1579702/0002/000`) to explicit state transitions.

Reading order. Chapter 1 situates the work and names the 234-entity scope; Chapter 2 defines the normalised record types; Chapter 3 covers the AI pipeline; Chapter 4 covers the variance and commentary engine; Chapter 5 covers the workflow and approval engine; Chapter 6 specifies controls, testing, and acceptance; Chapter 7 is a traceability appendix linking every requirement to a source document and a structured record in `structured/`.

Conventions. Workflow state identifiers appear in SMALL CAPS, for example CHECK_MATERIAL_VARIANCES. File paths are typeset in `monospace` and are relative to the repository root. Enum values in running text also appear in SMALL CAPS; they are defined authoritatively in `structured/enums.yaml`. Month-end-relative timing is expressed as `M±n` where `n` is the number of working days from period close.

Scope exclusion. SAP / PeopleSoft integration detail and the BSS Risk Assessment (a sibling process covered by a separate business case) are *not* specified here. The BSS RA source document is retained in the corpus for context only.

Chapter 1

Overview and Scope

1.1 Problem statement

Month-end Trial Balance review at SCB is currently executed as a spreadsheet-driven process. For each of the 234 Legal Entities (LEs) in the Group, a GFS Analyst rolls the prior month working file forward, downloads the PSGL master account extract, runs variance calculations in an EUC-governed Excel workbook, chases country stakeholders for commentary, tracks unposted journals, and hands a signed-off summary to the Head of Africa GFS for inclusion in the monthly FORTM pack. The cycle runs from `M-3` (preparation) to `M+7` (cycle close), with the critical review phase compressed into `M-2` through `M+5`.

Three problems follow from the current model:

1. **Effort.** The process consumes 20–25 FTEs at roughly four hours/day during the active review window — an establishment the business case identifies as the largest single automation candidate in the close calendar.
2. **Control surface.** Three EUC references govern the process (`EX/200/1353107/0001/000`, variance file `EX/200/1579702/0001/000`, and review control `EX/200/1579702/0002/000`). EUCs carry recurring evidencing burden and are misaligned with the UK ACG ambition to eliminate end-user spreadsheets.
3. **Latency.** The monthly batch cadence rules out daily monitoring; anomalies surface at month-end rather than when they arise.

1.2 In-scope workstreams

This specification covers the three workstreams confirmed with the business sponsor:

A. Canonical corpus. A machine-readable representation of every artefact under review — the 28-state BPMN workflow, five decision points, three EUC controls, six KPIs, eight process steps, and the domain glossary — normalised to JSON and validated against Draft-07 schemas.

B. AI augmentation. An on-premise pipeline hosted on the `vllm-only` policy (`data-product.routing = vllm-only`) that ingests the PSGL / S/4 extract, applies *statistical* anomaly detection (percentile-based thresholds, configurable per account category) at caption, segment, and LE levels, drafts commentary using historical patterns, and emits the FORTM narrative. Machine-learning anomaly detectors (Isolation Forest, autoencoder, Prophet) are scoped for v2.0 (Chapter 10) and are not part of the v1.0 deliverable.

C. Workflow and approval engine. A formalisation of the seven DOI activities and the gateway logic that decides when material variances require commentary, when journals must be posted, when further investigation is needed, and whether the entry-confirmation deadline at `M+4` has been honoured.

1.3 Out-of-scope

The following are referenced but not specified:

- Country IFRS reporting — handled by local Finance teams.
- BSS Risk Assessment — sibling process with its own business case. Its source document is retained in the corpus for terminology reuse only.
- SAP / PeopleSoft integration detail — the AI layer reads data through the same `SC Bridge` interface as the manual process; interface engineering is a separate document.
- Country expansion beyond the Hong Kong (HK) pilot — the HKG_TB and HKG_PL workbooks are the source of truth for the initial rollout; LE coverage extends to all 234 entities in phase 4 onward.

1.4 Stakeholders

Table 1.1 enumerates the stakeholders named in the business case and the DOI.

Table 1.1: Stakeholders and responsibilities.

Role	**Named**	**Responsibilities**
Process Owner	Modi, Piyush	BPMN ownership; workflow changes
Head of Africa GFS	—	Internal review; FORTM sign-off
GFS Analyst	—	Primary executor; variance file owner
Country Stakeholders	group	Commentaries; `M+4` entry confirmation
GCFO Teams	—	Consume narrative; decision-making
Group Team	—	FORTM recipient; further review
AI Architect	—	PSGL + S/4 compatibility
PPE / AI Architect	—	SAP integration assessment

1.5 Timeline

The process obeys a fixed month-end-relative timeline (Table 1.2). All state transitions in Chapter 5 carry a `timing` field that maps to one of these phases.

Table 1.2: Month-end-relative timeline phases.

Phase	Timing	Activities
Preparation	M-3	Roll forward; PSGL download; master refresh; parameter set
Extraction	M-3 to M+5	SC Bridge extract into the variance file (`EUC-002`)
Analysis	M-2 to M+5	Variance calc; commentary; investigation; risk
Review	M+5	Internal review (`EUC-003`); FORTM submission

1.6 Context diagram

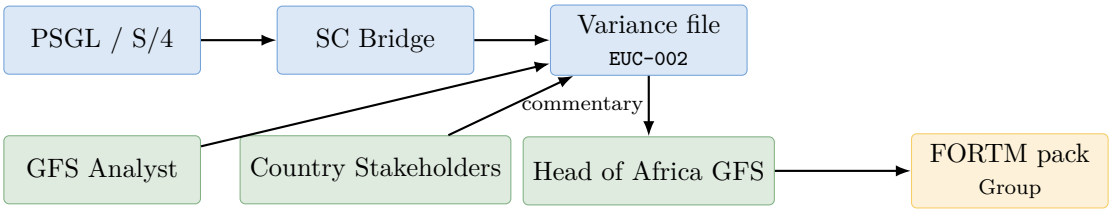

Figure 1.1: Context of the TB review process: three systems (blue), three human roles (green), and one downstream output (amber). The variance-analysis workbook is the primary EUC artefact.

1.7 Expected benefits

- FTE savings: 5 FTEs redeployed (business-case field `executiveSummary.fteImpact.expectedSavings`).
- Retirement of three EUCs (Group ACG alignment) on a phased schedule: the three Excel-based EUCs (`EUC-001`, `EUC-002`, `EUC-003`) are replaced by the rule-firing log plus the review dashboard audit trail (Chapter 8). Shared Drive evidence storage is retained only as a read-only archive for a one-year deprecation window (see Section 1.8); the dashboard audit trail is the authoritative evidence source from v1.0 cutover onwards.
- Enablement of daily controls (from monthly batch).
- Foundation for the SCB Finance LLM.

1.8 EUC retirement plan

The three EUCs are retired, not merely supplemented, on the following schedule. "Retirement" means the EUC is no longer the authoritative evidence for ACG sampling; the dashboard audit trail (Chapter 8, Section 8.5) is authoritative.

During the deprecation window (12 months from v1.0 go-live), the Shared Drive evidence folder is mirrored automatically from the dashboard export at month-end (read-only, with

Table 1.3: EUC retirement schedule.

EUC	Artefact	Retirement cutover	Successor evidence
EUC-001	PSGL master extract	Month of v1.0 go-live	Dashboard "Master Data" tab; SHA-256 of extract in audit trail
EUC-002	Variance analysis workbook	Month of v1.0 go-live	Dashboard "Variance Review" tab; per-item decision record
EUC-003	FORTM Pack	Month of v1.0 go-live + 1 cycle	Dashboard "FORTM Export" with signed PDF snapshot

mismatch alerting). After the deprecation window closes, the Shared Drive location is decommissioned and removed from the ACG sampling procedure. Section 6.3 in Chapter 6 is superseded by this plan for all v1.0 and later cycles; the Chapter 6 text is retained as a historical reference only.

Chapter 2

Canonical Data Schema

2.1 Overview

Every artefact under review is represented by a JSON record validated against a Draft-07 JSON Schema. Six schemas are defined; collectively they constitute the data contract consumed by the AI pipeline (Chapter 3) and by the controls engine (Chapter 4).

Table 2.1: Schemas under `structured/schema/`.

Schema	Validates	Records
`control.schema.json`	EUC and process controls	3
`decision-point.schema.json`	Gateway logic records	5
`workflow.schema.json`	State machine for a TB workflow	1
`requirement.schema.json`	Business-case requirement	1
`kpi.schema.json`	KPI definitions (collection file)	6
`glossary.schema.json`	Domain glossary	1 (28 terms)
`data-product.schema.json`	ODPS 4.1 data-product manifest	1

2.2 Control record

A `control` binds a governance reference (the EUC identifier) to one or more workflow states. It declares type, frequency, owner, and the evidence location examined at audit.

2.3 Decision-point record

A decision point encodes an exclusive gateway: an input metric, an operator, a threshold (where applicable), and an enumerated list of outcomes. Each outcome pairs a human-readable condition with a machine-actionable `action` verb. The `aiAugmentation` block names the substitution target for the manual judgement.

Table 2.2: `control.schema.json` fields.

Field	Type	Req	Notes
controlId	string	Y	`EUC-`*`nnn`*
eucReference	string	N	`EX/nnn/nnnnnnn/nnnn/nnn` pattern
controlObjective	string	Y	Plain-language objective
controlType	enum	Y	PREVENTIVE, DETECTIVE, CORRECTIVE
frequency	enum	Y	DAILY..ANNUALLY
timing	string	N	`M`±`n` phase
owner	string	Y	GFS role or named individual
evidenceLocation	string	N	Where audit evidence lives
relatedStates	array	Y	Workflow `stateIds` (≥1)
workflowRef	string	N	Workflow `workflowId`
source_document	object	N	Path, optional page and sha256

Listing 2.1: `EUC-002` variance-file control record (abbreviated).

```
{
  "controlId": "EUC-002",
  "eucReference": "EX/200/1579702/0001/000",
  "controlObjective": "TB Variance Analysis file integrity and calculation accuracy
      ",
  "controlType": "detective",
  "frequency": "monthly",
  "timing": "M-2 to M+5",
  "owner": "GFS Analyst",
  "evidenceLocation": "Shared Drive",
  "relatedStates": ["UPDATE_TB_DETAILS", "CALCULATE_VARIANCE", "FILTER_VARIANCES"],
  "description": "Ensures variance calculations in the Excel working file are
      accurate."
}
```

Listing 2.2: `DP-001` material variance gateway.

```
{
  "id": "DP-001",
  "name": "Material Variance Check",
  "stateRef": "CHECK_MATERIAL_VARIANCES",
  "inputMetric": "variance_amount",
  "operator": "exceeds_threshold",
  "threshold": "materiality_limit",
  "outcomes": [
    { "condition": "Variances Are Material", "action": "seek_commentaries" },
    { "condition": "Variances Not Material", "action": "proceed_to_unexplained_check
        " }
  ],
  "aiAugmentation": {
    "opportunity": "Auto-classify variance materiality using ML threshold tuning.",
    "dataNeeded": ["variance_amount", "account_category", "historical_thresholds"]
  }
}
```

2.4 Workflow record

The workflow record declares a state machine. Valid `type` values are START_EVENT, END_EVENT, TASK, GATEWAY_EXCLUSIVE, GATEWAY_PARALLEL, and INTERMEDIATE_EVENT. Each state carries optional `systems`, `inputs`, `outputs`, a `participant`, and a list of `transitions`. A transition in a gateway state additionally carries a `condition` string that matches one of the outcome labels in the associated decision point.

Table 2.3: Workflow state-machine header fields.

Field	Type	Req	Notes
workflowId	string	Y	Canonical slug (`tb-review-glo`)
version	string	Y	SemVer
title	string	Y	Long name
processOwner	string	N	Named owner
timeline	array	N	Phase / timing / description
states	array	Y	≥ 2 states, including one START_EVENT and one END_EVENT
dataObjects	array	N	Workflow-scoped artefacts
participants	array	N	Roles and groups

2.5 Requirement record

The requirement record captures the business case. It is scoped by a status enum (DRAFT, REVIEW, APPROVED, REJECTED, SUPERSEDED), contains executive summary, scope, stakeholders, process steps (each with an `aiPriority` enum from LOW..CRITICAL), decision points, KPI references, control references, risk assessment, and a phased implementation plan. The single record populated here is `TB-REQ-001` in state APPROVED.

2.6 KPI record

KPIs are stored as a collection file at `structured/kpis/tb-kpis.json` containing one array entry per KPI (Table 2.4). Each entry is validated against `kpi.schema.json`; the wrapper is checked by `structured/validate.py`.

Table 2.4: The six KPIs defined for the TB process.

KPI	Type	Description	Target
TB-KPI-001	threshold	Variance materiality threshold	country-specific
TB-KPI-002	duration	TB review cycle time	reduce via daily controls
TB-KPI-003	percentage	Commentary coverage	100
TB-KPI-004	effort	FTE effort	15–20 FTEs
TB-KPI-005	percentage	Unexplained variance rate	0
TB-KPI-006	percentage	EUC control compliance	100

2.7 Glossary record

One glossary record contains 28 terms drawn verbatim from the business case and the DOI. Category membership (BALANCE_SHEET, INCOME_STATEMENT, GENERAL, REGULATORY) is used by the GlossaryService pair-type alias expansion at query time.

2.8 Data-product record

The data-product record is an ODPS 4.1-conformant manifest. It names the routing policy (`vllm-only`), the data security class (CONFIDENTIAL), the owner, the per-country view for HK, the system prompt, and the RAG source wiring. Consumers read this record to know whether a given user request is permitted to contact external LLM providers — the answer for all TB artefacts is *no*.

2.9 Enumerations

All enumerated values used in any schema or record are listed authoritatively in `structured/enums.yaml`. Table 2.5 summarises the most frequently referenced groups.

Table 2.5: Frequently referenced enumerations.

Enum	**Values**
control_type	preventive, detective, corrective
workflow_state_type	start_event, end_event, task, gateway_exclusive, gateway_parallel, intermediate_event
decision_operator	equals, not_equals, exceeds_threshold, within_threshold, in_set, boolean
ai_priority	low, medium, high, critical
risk_severity	L, M, H
kpi_measure_type	threshold, duration, percentage, effort, count, ratio
data_security_class	public, internal, confidential, restricted

2.10 Validation

A local registry-based validator is provided at `structured/validate.py`. It walks the record directories, maps each to its schema via `RECORD_DIRS`, and reports per-file results. The script is offline: it instantiates a `referencing.Registry` loaded from `structured/schema/` so no network resolution is attempted.

Listing 2.3: Running the validator.

```
$ python3 docs/tb/structured/validate.py
[ok]    controls/euc-001-tb-data-extraction.json
[ok]    controls/euc-002-variance-analysis-file.json
[ok]    controls/euc-003-internal-review.json
[ok]    decision-points/dp-001-material-variance.json
...
```

```
[ok]   corpus.json (17 items, all paths resolved)
13/13 records validated, corpus OK
```

2.11 Unified Schema Registry

As of 18 April 2026, all schemas in this specification are published in the unified schema registry at `docs/schema/`. Migration status is *complete* (see `docs/schema/registry.json` field `migration.status`). This consolidation addresses cross-document schema divergence identified in the specification review.

Registry index. `docs/schema/registry.json` — Master index of all schemas across domains.

Common types. `docs/schema/common/base-types.schema.json` — Shared definitions (`Metadata`, `Currency`, `DateRange`) referenced via JSON Schema `$ref`.

Consolidated enums. `docs/schema/common/enums.yaml` — Single source of truth for enumeration values including `account_category`, `variance_flag`, `decision_point`, and `workflow_status`.

TB schemas. `docs/schema/tb/` — Domain-specific schemas: `tb-extract.schema.json`, `pl-extract.schema.json`, `variance-record.schema.json`, `commentary-draft.schema.json`, and `decision-point.schema.json`. All marked `status: stable`.

The legacy YAML schemas at `docs/tb/machine-readable/training-data/hkg-tb-schema.yaml` and `docs/tb/machine-readable/training-data/hkg-pl-schema.yaml` have been converted to JSON Schema Draft-07 in the unified registry; the registry entries record the `legacy_path` for audit traceability. The legacy files are retained read-only during a one-release deprecation window and will be removed in v1.1. Validation:

```
python3 docs/schema/validate.py --registry-check
python3 docs/schema/validate.py --domain tb \
    --schema tb-extract.schema.json --file tb-hkg-2026-03.json
```

CI runs `--registry-check` on every pull request that touches `docs/schema/` or any TB pipeline module.

Chapter 3

AI Augmentation Pipeline

3.1 Architecture

The AI pipeline is a five-stage request-response chain, hosted on the internal vLLM cluster per the `vllm-only` routing policy in `docs/tb/structured/data-products/trial-balance-review-v1.json`. Every stage reads and writes records validated by Chapter 2; no blob passes between stages without a schema binding.

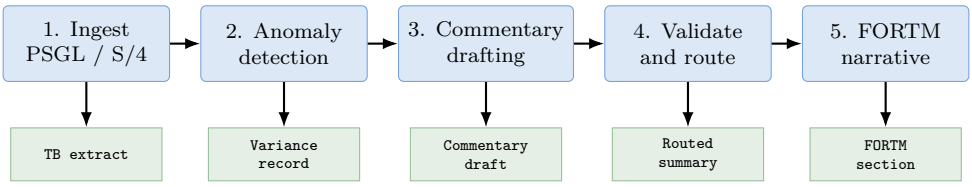

Figure 3.1: Five-stage AI pipeline; each stage emits a schema-bound artefact.

3.2 Stage 1: Ingestion

Input. PSGL master account data and the SC Bridge TB extract, one per LE, keyed on the `UPDATE_TB_DETAILS` workflow state.

Transform. Column normalisation against `docs/tb/machine-readable/training-data/hkg-tb-schema.yaml` (13 sheets). Bank CSV samples live in `docs/tb/extracted-text/hkg-tb-sample.csv` and `docs/tb/extracted-text/hkg-pl-sample.csv`.

Acceptance. `EUC-001` evidence rules apply: the extraction must preserve debit/credit totals and every row must carry a valid account number from the refreshed master.

3.3 Stage 2: Anomaly detection

The anomaly detector performs four orthogonal checks per account (Table 3.1). Thresholds are account-category-sensitive and are re-fit monthly from the prior twelve months; the fitted thresholds are persisted to the variance file EUC as audit evidence.

Table 3.1: Anomaly checks run per account.

Check	Metric	Rule
Period spike	$\lvert\Delta\rvert$	$> 3\sigma$ over prior 12 months[†]
Sign flip	$\text{sign}(v_t)$	$\neq \text{sign}(v_{t-1})$ and $\lvert v_t\rvert >$ threshold
New account	existence	account absent from prior-period master
Stale balance	$\lvert\Delta\rvert$	$= 0$ for ≥ 3 consecutive periods with prior activity

[†]**Assumption caveat:** The 3σ threshold assumes normally-distributed variance values. This assumption has not been statistically validated against the TB population; financial data often exhibits fat tails (leptokurtosis). The pilot will include a Shapiro-Wilk normality test per account category. Non-normal distributions may require robust alternatives (e.g., MAD-based z-scores or quantile thresholds). Thresholds marked "TBD pending pilot."

Only accounts flagged by at least one check enter Stage 3. Accounts below materiality (`DP-001`) are still logged for audit but short-circuit to Stage 4.

3.4 Stage 3: Commentary drafting

Model. A vLLM-hosted instruction-tuned model accessed via the internal gateway. The system prompt is the `x-prompting-policy.systemPrompt` field of the data product record; `maxTokens = 4096`, `temperature = 0.3`, `responseFormat = "text"`.

Prompt structure. Each commentary request carries:

- The flagged variance record (account, prior, current, delta).
- The three most similar historical commentaries retrieved from the `TB_REVIEW_VECTORS` table (55 chunks, 220-word chunking with 40-word overlap; see `machine-readable/rag/rag_embedding_records.jsonl`).
- The account's finance caption and segment.
- Any related journal identifiers found in the GL for the period.

Output contract. The draft is returned as a `commentary_draft` blob that is paired with the variance record before leaving the stage. Human review by the GFS Analyst is mandatory (per DOI `REVIEW_INCORPORATE`); the draft is a *suggestion*, never a final artefact.

3.5 Stage 4: Validation and routing

The gateway logic defined in Chapter 5 is evaluated here. Table 3.2 maps each decision point to the record fields the engine reads.

Table 3.2: Decision-point routing.

DP	Gateway state	Read fields
DP-001	CHECK_MATERIAL_VARIANCES	variance_amount, account_category, materiality_limit
DP-002	IS_JOURNAL_POSTED	journal_status, posting_date
DP-003	CHECK_FURTHER_INVESTIGATION	investigation_status, entry_age_days
DP-004	CHECK_CONFIRMATION	confirmation_received, days_to_deadline (M+4)
DP-005	CHECK_UNEXPLAINED	unexplained_variance_count

3.6 Stage 5: FORTM narrative generation

The final stage composes the FORTM pack section. It consumes the routed summary plus the KPI telemetry (Table 2.4) and emits a structured narrative containing: executive summary, material variance table, unexplained items, risk register update, and the list of outstanding journal postings. The narrative goes to REVIEW_INCORPORATE for `EUC-003` sign-off before SEND_FORTM.

3.7 Controls posture of the pipeline

Every stage is in-scope of the UK ACG automated-controls regime. The per-stage control posture is summarised in Table 3.3.

Table 3.3: Per-stage controls posture.

Stage	Control type	Mechanism
Ingest	preventive	Totals reconciliation vs PSGL source
Anomaly	detective	Threshold log + human sampling
Commentary	detective	Mandatory human review (DOI)
Validation	preventive	Schema + gateway enforcement
FORTM	detective	EUC-003 senior sign-off

3.8 Evaluation harness

Golden set. The HK Nov 2025 workbooks (`source/HKG_TB review Nov'25.xlsm` and `source/HKG_PL review Nov'25.xlsm`) serve as regression data. All 234 LEs enter at phase 4 of the implementation plan.

Commentary metric. Draft quality is scored against analyst accepts in the DOI review phase; target accept-without-edit $\geq 70\%$, accept-with-edit $\geq 95\%$.

Anomaly metric. Per-check precision/recall against analyst flags in a held-out period.

Cycle time. Measured as `TB-KPI-002`; target is the enablement of a daily cadence, which collapses the cycle from `M+1..M+7` to a rolling one-day delay.

Chapter 4

Controls and Commentary Engine

4.1 Role of the engine

The controls engine is the declarative layer that turns the five decision-point records of Section 2.3 into executable rules. It sits between Stage 2 (anomaly detection) and Stage 5 (FORTM narrative) of the AI pipeline. Its inputs are the variance records, the GL journal extract, and the stakeholder commentary responses; its outputs are routing actions and a commentary register.

4.2 Decision tree

Figure 4.1 collapses the five gateways into a single cascade. The tree is identical to the workflow in Chapter 5 but omits task nodes; each leaf is a terminal action in the gateway vocabulary.

4.3 Rule format

Each rule is a record with five fields:

Listing 4.1: Generalised rule form.

```
{
  "id": "<DP-nnn>",
  "state": "<workflow stateId>",
  "input": "<record field path>",
  "operator": "<exceeds_threshold|boolean|in_set|...>",
  "threshold": "<value or parameter name>",
  "branches": [ {"condition":"...","action":"..."} ... ]
}
```

Rule evaluation is deterministic. When a record enters a gateway state, the engine reads the named input field, applies the operator, and dispatches the matching branch's action.

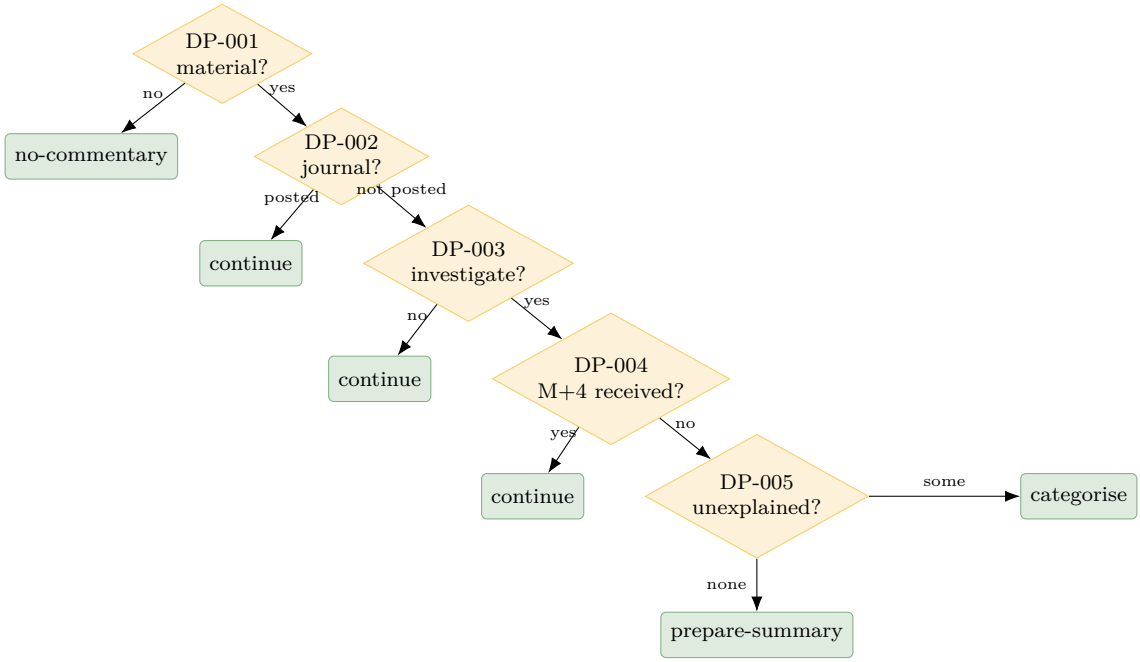

Figure 4.1: Decision cascade encoded by the five decision-point records.

Actions reference downstream tasks in the workflow (Chapter 5).

4.4 Materiality thresholds

`DP-001` reads `materiality_limit` from the country parameters sheet. During the AI rollout, the fitted ML threshold (Stage 2) is compared side-by-side with the manual limit for six months; divergences $> 10\%$ surface to the Head of Africa GFS for review. Only after six clean cycles does the fitted threshold replace the manual one in the parameter sheet.

4.5 Journal posting

`DP-002` has three branches. The engine first reads `journal_status` from the GL extract. If the status is POSTED, the routing short-circuits to CHECK_UNEXPLAINED. If POSTING_REQUIRED, the engine emits a posting request via SEND_JOURNAL_REQUEST and returns to CHECK_FURTHER_INVESTIGATION. If NOT_POSTED (stuck), the path proceeds directly to CHECK_FURTHER_INVESTIGATION, skipping the request.

4.6 Investigation and deadlines

`DP-003` is a boolean gateway controlled by the `investigation_status` field. `DP-004` enforces the `M+4` deadline by reading `days_to_deadline` from the variance record. Crossing the deadline triggers the TRACK_ENTRIES branch and the risk-assessment update in UPDATE_RISK.

4.7 Unexplained-variance handling

`DP-005` is the final gate before summary. If `unexplained_variance_count` = 0, the engine transitions to PREPARE_SUMMARY. Otherwise, the unexplained items are passed to CATEGORIZE_UNEXPLAINED, where each item is tagged against the BSS risk vocabulary (SUBSTANTIATED AT RISK, UNSUBSTANTIATED UNOWNED, UNSUBSTANTIATED OWNED) and a commentary draft is requested.

4.8 Commentary register

All drafts produced by Stage 3 of the AI pipeline, together with the analyst accept/edit decision, are persisted to the commentary register. The register is the primary evidence artefact for `EUC-002`: the register row count must equal the flagged variance row count at the end of UPDATE_VARIANCE_COMMENTARIES.

Table 4.1: Commentary register schema (derived from the variance record).

Field	Type	Notes
account_id	string	GL account number
period	string	ISO yyyy-mm
variance_amount	number	Prior → current delta
draft	string	LLM-generated narrative
final	string	Post-review narrative
accept_mode	enum	AS-IS, EDITED, REJECTED
reviewer	string	GFS Analyst name or role
review_ts	datetime	ISO-8601

4.9 Per-country parameter views

The data product record carries a `x-country-views.HK` block that injects an HKMA-aware vocabulary and sets `defaultFilters.entity = "HKG"`. Adding a new country is a matter of adding another view; no schema change is needed. Until additional views exist, Hong Kong is the only LE with a fitted threshold history.

4.9.1 Parameter loading and validation for 234 entities

Full Group rollout to all 234 Legal Entities (LEs) is gated on the availability of a per-entity parameter file, not on model capacity. This section specifies how those parameters are loaded, validated, and versioned so that entities can be on-boarded without schema or code changes.

Parameter file layout. Per-entity parameters live at `docs/tb/machine-readable/entity-params/<le-code>.yaml` (one file per LE; `le-code` matches the three-letter ISO-ish LE code used by PSGL, e.g. `HKG`, `SGP`). Each file is a dictionary with keys `materiality_threshold`, `variance_bands`, `account_category_overrides`, `fx_reporting_currency`, `regulator_vocabulary`, `commentary_style_exemplars`, and `last_calibration_date`.

Schema. All entity parameter files validate against `docs/schema/tb/entity-params.schema.json` (new, registered in `docs/schema/registry.json` under the `tb` domain). The CI `--registry-check` command fails if any file deviates.

Loading. The TB pipeline loads parameters at batch start via `EntityParamsRegistry.load_all()` (`src/tb/entity_params.py`) which reads every file in the directory, validates it against the schema, and caches the result. Missing LE codes are logged as `warn`; a missing code does not block the pipeline for *other* entities, but processing for the missing LE is skipped with an audit record.

Onboarding a new LE. The runbook is `docs/tb/runbooks/onboard-le.md`: (1) copy `HKG.yaml` as a template, (2) fit a threshold history against 12 months of TB data, (3) open a pull request with the new file, (4) CI validates the file and runs a dry-run cycle on the new LE, (5) reviewer sign-off (senior analyst + controller).

Roll-forward safety. Parameter files are versioned in git; the data product record records the git SHA of the entity-params directory at run start (`runMetadata.entityParamsSha`). Retrospective ACG queries reconstruct parameters at the historical run time.

Rollout readiness (see Chapter 1, Section on Timeline): HKG is fully onboarded; a wave of 8–10 LEs is added in phase 4 (quarter after v1.0 go-live), with the remaining 220+ onboarded over 12–18 months. The dashboard (Chapter 8) surfaces per-LE coverage so that operations can prioritise LEs by review volume and regulator risk.

4.10 Observability

Every rule firing is logged with input, branch taken, timestamp, and the version of the ruleset. Logs are aggregated into the `TB_REVIEW_VECTORS` table's metadata index so that prior decisions are retrievable by account and period.

Chapter 5

Workflow and Approval Engine

5.1 Source

The workflow engine formalises the 28-state BPMN diagram captured in `structured/workflows/tb-review-glo.json`. The diagram is derived from two source artefacts:

- `source/Operation of Month End Close Controls - Trial Balance Review - GLO (1).pdf` — the control narrative that defines the seven DOI activities,
- `source/Business case - Trial Balance (2).doc` — the business case that expands the activities into the eight process steps of `TB-REQ-001`.

5.2 State inventory

Table 5.1 enumerates the 28 states. Tasks are numbered sequentially; gateways and events are marked in the **Type** column.

5.3 Preparation phase (M-3)

Figure 5.1 shows states 1–6. Execution is strictly sequential; every state has exactly one outgoing transition.

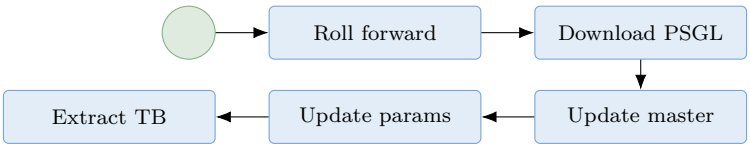

Figure 5.1: Preparation phase (states 1–6). EUC-001 governs the PSGL download and TB extraction steps.

Table 5.1: States in `tb-review-glo` (28 in total).

#	State ID	Type	Timing	Name
1	START	start	—	Prepare Schedule Work Day
2	ROLL_FORWARD	task	M-3	Roll Forward Prior Month File
3	DOWNLOAD_PSGL	task	M-3	Download PSGL Master Account File
4	UPDATE_ACCOUNTS_MASTER	task	M-3	Update Accounts Master Sheet
5	UPDATE_PARAMETERS	task	M-3	Update Parameters and Comparatives
6	EXTRACT_TB	task	M-3..M+5	Extract Trial Balance
7	UPDATE_TB_DETAILS	task	M-2..M+5	Update TB Details (EUC-002)
8	CALCULATE_VARIANCE	task	M-2..M+5	Calculate Variance
9	FILTER_VARIANCES	task	M-2..M+5	Filter Variances
10	CHECK_MATERIAL_VARIANCES	gw-excl	M-2..M+5	Material variance? (DP-001)
11	SEEK_COMMENTARIES	task	M-2..M+5	Seek Commentaries
12	RECEIVE_COMMENTARIES	intermediate	M-2..M+5	Receive Commentaries
13	UPDATE_COMMENTARIES	task	M-2..M+5	Update Commentaries
14	UPDATE_VARIANCE_COMMENTARIES	task	M-2..M+5	Finalise Commentaries
15	IS_JOURNAL_POSTED	gw-excl	M-2..M+5	Journal posted? (DP-002)
16	SEND_JOURNAL_REQUEST	task	M-2..M+5	Send Posting Request
17	CHECK_FURTHER_INVESTIGATION	gw-excl	M-2..M+5	Investigate? (DP-003)
18	CHECK_ENTRIES_STATUS	task	M-2..M+5	Check with Team
19	CHECK_CONFIRMATION	gw-excl	M-2..M+5	M+4 received? (DP-004)
20	TRACK_ENTRIES	task	M-2..M+5	Track and Risk Analyse
21	UPDATE_RISK	task	M-2..M+5	Update Risk Assessment
22	CHECK_UNEXPLAINED	gw-excl	M-2..M+5	Unexplained? (DP-005)
23	CATEGORIZE_UNEXPLAINED	task	M-2..M+5	Categorise Unexplained
24	PREPARE_SUMMARY	task	M-2..M+5	Prepare Summary
25	SEND_INTERNAL_REVIEW	task	M+5	Send for Internal Review
26	REVIEW_INCORPORATE	task	M+5	Review (EUC-003)
27	SEND_FORTM	task	M+5	Send FORTM Summary
28	END	end	—	TB Review Completed

5.4 Analysis phase (M-2 to M+5)

The analysis phase is the control-rich core of the process. It holds all five gateways, all commentary handling, and the journal-posting cycle. Figure 5.2 shows states 7–24.

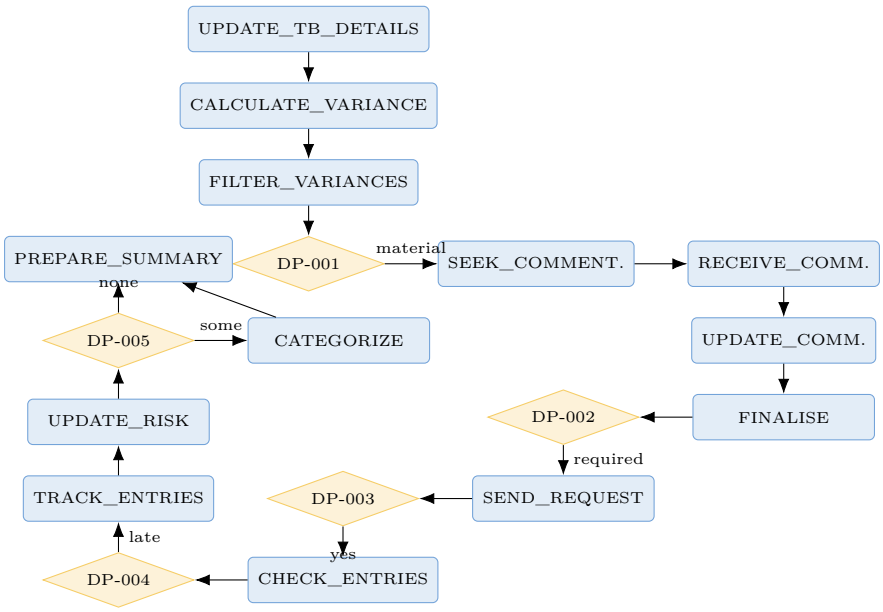

Figure 5.2: Analysis phase. Gateways DP-001..DP-005 are diamond nodes; commentary loop sits to the right of DP-001; journal-posting loop sits below DP-002. Not all transitions are drawn; see the full JSON for completeness.

5.5 Review phase (M+5)

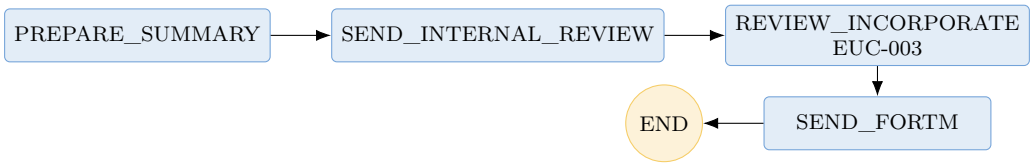

Figure 5.3: Review phase: senior sign-off (EUC-003) gates the FORTM submission.

5.6 Approval matrix

Table 5.2: Approval and review responsibilities.

State	Participant	Decision	Control
UPDATE_TB_DETAILS	GFS Analyst	File integrity	EUC-002
UPDATE_COMMENTARIES	GFS Analyst	Draft accept/edit/reject	—
REVIEW_INCORPORATE	Head of Africa GFS	Senior sign-off	EUC-003
SEND_FORTM	Head of Africa GFS	Release to Group	—

5.7 Data objects

The five data objects declared on the workflow are listed in Table 5.3.

Table 5.3: Workflow data objects.

ID	Name	Description	EUC
DO-001	Variance Analysis File	Primary working file	EUC-002
DO-002	Shared Drive	Evidence store	—
DO-003	PSGL Master Account File	Master data extract	EUC-001
DO-004	SC Bridge	Extract interface	—
DO-005	FORTM Pack	Group-level reporting package	EUC-003

5.8 Exception paths

The workflow diagram encodes the happy path and the four gateway-driven loops. Two operational exceptions are not drawn but are handled by the engine:

1. **SC Bridge failure.** If EXTRACT_TB cannot read from the bridge, the state is parked and a human notification is emitted. Re-entry resumes from EXTRACT_TB; earlier states are not replayed.
2. **Late FORTM change.** If Group requests a revision after END, a new run is created with a suffix `-revN`; evidence for the original run is retained.

Chapter 6

Controls, Testing, and Acceptance

6.1 Control enforcement map

Figure 6.1 binds each EUC control to the workflow states it governs. The binding is authoritative: the `relatedStates` array of each control record is the source of truth.

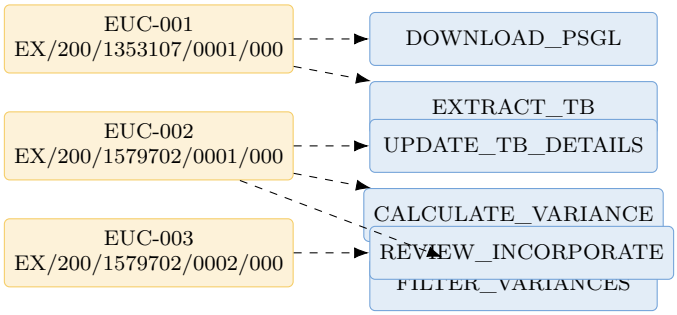

Figure 6.1: Control-to-state binding. Dashed lines denote the `relatedStates` relation.

6.2 Risk register

Table 6.1 lists the four control risks enumerated in `docs/tb/machine-readable/process/tb-review-controls.yaml` and the two business-case risks in `docs/tb/structured/requirements/tb-business-requirements.json`.

6.3 Test strategy

Testing is layered; each layer has a named artefact.

Schema tests. `structured/validate.py` runs over every record on CI. Exit code non-zero on validation failure; the script also verifies that every `source_path` and `text_path` in `corpus.json` resolves.

Rule tests. A fixture set lives at `machine-readable/training-data/` (two CSVs are checked in as the HK golden sample). Every rule in Chapter 4 is exercised against

Table 6.1: Risks and mitigations.

ID	Severity	Description	Today	AI mitigation
CR-001	high	Manual extraction, EUC risk	EUCs 001/002	Fully automated ingest
CR-002	medium	Late commentary responses	M+4 chasing	Auto-drafts remove dependency
CR-003	high	Incomplete analysis under time pressure	20–25 FTEs	Parallel anomaly detection
CR-004	medium	Not scalable to daily ops	none	Foundation for daily cadence
TB-RISK-001	H	PSGL-only coverage today	PSGL adapters only	Build S/4 adapter
TB-RISK-002	H	SAP integration dependency	PPE review	Compatibility assessment

at least one positive and one negative fixture row; the expected branch is asserted.

Workflow tests. The state machine is walked dry in CI: every transition in `tb-review-glo.json` must reach END in at least one simulated path.

Commentary quality tests. The accept-without-edit rate (Chapter 3) is tracked as a monthly KPI. Drops > 5 pp month-on-month raise a governance ticket.

EUC evidencing. *Superseded by the EUC retirement plan* (Section 1.8 in Chapter 1). From v1.0 go-live onwards, the dashboard audit trail (Chapter 8, Section 8.5) is the authoritative evidence for ACG sampling. During the 12-month deprecation window, the Shared Drive mirror of the dashboard export is retained read-only; ACG samplers may consume either source during that window, but the dashboard audit trail is normative if the two diverge. The historical text of this paragraph ("For each of `EUC-001`, `EUC-002`, `EUC-003`, the evidence folder on Shared Drive must contain the rule-firing log and the reviewer signature") applied to pre-v1.0 cycles only.

6.4 Acceptance criteria

1. All 13 structured records validate against their schemas (Chapter 2, Listing 2.3).
2. The HK golden workbook produces a draft FORTM section that the Head of Africa GFS accepts with $\leq 20\%$ of the material-variance commentaries edited.
3. The cycle time from EXTRACT_TB to SEND_FORTM on the HK workbook is ≤ 5 working days (parity with today, a precondition for extending to daily cadence).
4. `TB-KPI-005` (unexplained variance rate) is zero at SEND_FORTM for the HK golden run.
5. ACG evidence for all three EUCs is reproducible by the rule log alone, without access to the underlying Excel workbook.

6.5 Deployment and rollback

Table 6.2 specifies the sign-off matrix and rollback behaviour for each environment.

Table 6.2: Deployment sign-off and rollback.

Env	Sign-off	Rollback	Notes
DEV	AI Engineer	force-push	Latest green only.
UAT	Process Owner	blue/green	HK workbook reproduces on each deploy.
PROD	Head of Africa GFS	blue/green	New ruleset runs parallel to manual for one close cycle before switchover.

6.6 Training and change management

The 20–25 FTEs currently executing the process are trained as exception handlers and rule reviewers. The operating model shifts from *analyst produces commentary* to *analyst accepts, edits, or rejects a draft.* Accept/edit telemetry is used to retrain the draft-generation prompts once per quarter.

6.7 Traceability matrix

Table 6.3 links every numbered process step from `TB-REQ-001` to the workflow states and controls that realise it.

Table 6.3: Process step traceability.

Step	Name	Priority	Realised by
TB-STEP-001	Roll Forward Prior Month File	medium	ROLL_FORWARD
TB-STEP-002	Download PSGL, update master data	high	DOWNLOAD_PSGL, UPDATE_ACCOUNTS_MASTER, UPDATE_PARAMETERS (EUC-001)
TB-STEP-003	Extract Trial Balance	critical	EXTRACT_TB (EUC-001)
TB-STEP-004	Calculate and filter variances	critical	UPDATE_TB_DETAILS, CALCULATE_VARIANCE, FILTER_VARIANCES, DP-001 (EUC-002)
TB-STEP-005	Seek and update commentaries	critical	SEEK_COMMENTARIES through UPDATE_VARIANCE_COMMENTARIES
TB-STEP-006	Journal posting and investigation	high	DP-002, SEND_JOURNAL_REQUEST, DP-003, CHECK_ENTRIES_STATUS
TB-STEP-007	Track entries and risk assessment	high	DP-004, TRACK_ENTRIES, UPDATE_RISK, DP-005, CATEGORIZE_UNEXPLAINED
TB-STEP-008	Prepare summary and internal review	critical	PREPARE_SUMMARY through SEND_FORTM (EUC-003)

Chapter 7

References

This appendix lists every file that backs a normative statement in the specification. Paths are relative to the repository root. SHA-256 hashes are taken from `structured/corpus.json` and are the ground truth for the provenance check of Chapter 2.

7.1 Source documents

Short name	Path	SHA-256 (prefix)
TB GLO control narrative	`docs/tb/source/OperationofMonthEndCloseControls-TrialBalanceReview-GLO(1).pdf`	`71a4524b9ec7…`
TB Business Case	`docs/tb/source/Businesscase-TrialBalance(2).doc`	`32744df8ea13…`
BSS RA Business Case	`docs/tb/source/Businesscase-BSSRiskAssessment(2).doc`	`dab433cecf59…`
DOI for TB Process	`docs/tb/source/DOIforTrialBalanceProcess.docx`	`b9d52d365b6b…`
HKG TB Workbook (Nov'25)	`docs/tb/source/HKG_TBreviewNov'25.xlsm`	`69f5da1f4c71…`
HKG PL Workbook (Nov'25)	`docs/tb/source/HKG_PLreviewNov'25.xlsm`	`47716e8c107b…`

7.2 Derived text

Extracted text for each source document lives at `extracted-text/`. The Word and PDF files are converted to Markdown; the macro-enabled workbooks produce CSV samples (one per book) that are sufficient to seed downstream joins without carrying the macro payload. Extraction uses `pandoc` for `.docx`, `pdftotext -layout` for born-digital PDFs, and a small Python pipeline for the `.xlsm` sheets.

7.3 Structured records

All of the following are JSON records validated by `structured/validate.py`.

Type	Path
Workflow	`structured/workflows/tb-review-glo.json`
Control	`structured/controls/euc-001-tb-data-extraction.json`
Control	`structured/controls/euc-002-variance-analysis-file.json`
Control	`structured/controls/euc-003-internal-review.json`
Decision point	`structured/decision-points/dp-001-material-variance.json`
Decision point	`structured/decision-points/dp-002-journal-posting.json`
Decision point	`structured/decision-points/dp-003-further-investigation.json`
Decision point	`structured/decision-points/dp-004-entry-confirmation.json`
Decision point	`structured/decision-points/dp-005-unexplained-variance.json`
Requirement	`structured/requirements/tb-business-requirements.json`
KPI collection	`structured/kpis/tb-kpis.json`
Glossary	`structured/glossary/tb-glossary.json`
Data product	`structured/data-products/trial-balance-review-v1.json`
Index	`structured/corpus.json`
Enumerations	`structured/enums.yaml`

7.4 Tools

Tool	Purpose
`pandoc`	`.docx` → Markdown.
`pdftotext` -layout	Text extraction from the GLO control PDF.
Python + `openpyxl`	Sample rows out of `.xlsm` workbooks.
`jq`	Query and cross-check `corpus.json`.
`jsonschema` / `referencing` (Python)	Offline schema validation with a local registry.
`xelatex` / `tectonic`	Build this document.

7.5 Minimal corpus excerpt

The entire corpus is in-tree under . For readers without access to the Git repository, Listing 7.1 reproduces the first three items of `structured/corpus.json` verbatim.

7.6 Out-of-scope assets

The BSS Risk Assessment business case (`docs/tb/source/Businesscase-BSSRiskAssessment(2).doc`) is kept in the corpus for terminology alignment but is out of scope for this specification; it is covered by a sibling business case with its own lifecycle.

Listing 7.1: First three items of `corpus.json`; identical SHA-256 values are reproduced in the table above.


```
{
  "schema_version": "1.0",
  "items": [
    {
      "id": "tb-glo-control-operation",
      "type": "control_narrative",
      "country": "GLO",
      "language": "en",
      "sha256": "71a4524b9ec7c6701a7137579ef6c8730eb53fcde2fe5a50515a4b870b560afc",
      "source_path": "docs/tb/source/Operation of Month End Close Controls ...GLO
          (1).pdf",
      "text_path": "docs/tb/extracted-text/bpmn-tb-review-glo.md",
      "structured_path": "docs/tb/structured/workflows/tb-review-glo.json"
    },
    {
      "id": "tb-business-case",
      "type": "business_case",
      "country": "GLO",
      "language": "en",
      "sha256": "32744df8ea13f12a36f6117f18874e578c627be6fbc3c48d2f73d32dcfa0c689",
      "source_path": "docs/tb/source/Business case - Trial Balance (2).doc",
      "text_path": "docs/tb/extracted-text/business-case-trial-balance.md",
      "structured_path": "docs/tb/structured/requirements/tb-business-requirements.
          json"
    },
    {
      "id": "bss-ra-business-case",
      "type": "business_case",
      "country": "GLO",
      "language": "en",
      "sha256": "dab433cecf598b908d070e2ebfd6f285565b7b3e1102cd2496e639dee1d0c28c",
      "source_path": "docs/tb/source/Business case - BSS Risk Assessment (2).doc"
    }
  ]
}
```

Chapter 8

Human Review Dashboard Specification

This chapter specifies the human-in-the-loop review interface, addressing the review finding that "human-in-the-loop is still a black box." The specification applies to both Trial Balance review and Arabic AP workflows.

8.1 Overview

The review dashboard provides a unified interface for human reviewers to:

1. View items queued for review (from AI-generated outputs requiring human validation)
2. Accept, edit, or reject AI-produced content
3. Track review metrics and team performance
4. Generate audit trails for compliance

8.2 Interface requirements

8.2.1 Queue view

The queue view displays all items pending human review.

Table 8.1: Queue view required fields.

Field	Type	Description
`item_id`	string	Unique identifier for the review item
`priority`	enum	CRITICAL, HIGH, NORMAL, LOW
`source_domain`	enum	TB, ARABIC, SIMULA
`queued_at`	datetime	When item entered queue
`due_by`	datetime	SLA deadline
`ai_confidence`	float	AI confidence score (0–1)
`estimated_time`	duration	Estimated review time
`assigned_to`	string?	Assigned reviewer (if any)
`status`	enum	PENDING, IN_PROGRESS, COMPLETED

Sorting and filtering

The queue **must** support:

- Sort by priority (default), due date, confidence, queued time
- Filter by domain, priority, status, assigned reviewer
- Search by item ID or related entity (invoice number, account code)

Bulk operations

For efficiency, the queue **must** support:

- Bulk assignment to reviewer
- Bulk priority change
- Bulk re-queue (return to pool)

8.2.2 Review view

The review view presents a single item for detailed examination.

Review View Wireframe

Source Document	**AI Extraction**
[Original image/text]	[Extracted fields with confidence]

Accept Edit Reject Escalate

Comment: _____

Figure 8.1: Conceptual wireframe for review view layout.

Required components

1. **Source panel**: Display original input (scanned image, OCR text, TB extract)
2. **AI output panel**: Display AI-generated content with field-level confidence indicators
3. **Comparison highlights**: Visual diff between source and extraction
4. **Action buttons**: Accept, Edit, Reject, Escalate, Skip
5. **Comment field**: Free-text for reviewer notes
6. **History panel**: Previous review actions on this item

Field-level editing

When reviewer selects "Edit":

- Each extracted field becomes editable

- Original AI value shown alongside edit field
- Change tracking records before/after values
- Validation runs on edited values before save

8.2.3 Analytics view

The analytics view provides management visibility into review operations.

Table 8.2: Required analytics metrics.

Metric	Aggregation	Purpose
Daily throughput	Count by day	Volume tracking
Average review time	Mean, p50, p95	Efficiency monitoring
Acceptance rate	Percentage	AI quality indicator
Edit rate by field	Percentage per field	Field-specific quality
SLA compliance	Percentage on-time	Operational health
Reviewer performance	By reviewer	Workload balancing
Escalation rate	Percentage	Training needs indicator

8.3 User workflow specification

8.3.1 Standard review flow

1. Reviewer logs in and views queue (filtered by assignment or domain)
2. Reviewer claims or is assigned an item
3. System displays review view with source and AI output
4. Reviewer examines content and selects action:
    - **Accept**: AI output is correct, proceed to next step
    - **Edit**: Make corrections, then accept
    - **Reject**: AI output is fundamentally wrong, return for reprocessing
    - **Escalate**: Beyond reviewer authority, route to supervisor
5. System records decision with timestamp and reviewer ID
6. Item moves to next workflow stage or back to queue

8.3.2 Escalation flow

Listing 8.1: Escalation routing rules.

```
if item.amount > reviewer.doa_limit:
    escalate_to = reviewer.supervisor
elif item.ai_confidence < 0.3:
    escalate_to = domain_expert
elif item.has_flag("regulatory_review"):
    escalate_to = compliance_team
else:
    escalate_to = team_lead
```

8.4 SLA specification

Table 8.3: Review SLA targets by priority.

Priority	Target	Definition
CRITICAL	4 hours	Regulatory deadline or high-value (>$1M)
HIGH	24 hours	Month-end close items
NORMAL	48 hours	Standard processing
LOW	72 hours	Training data or non-urgent

SLA countdown **must**:

- Exclude non-business hours (configurable by region)
- Pause when item is actively being reviewed
- Trigger alerts at 75% and 90% of time elapsed

8.5 Audit trail specification

Every user action **must** be logged to the audit trail.

8.5.1 Required audit events

Table 8.4: Audit events to be captured.

Event	Captured Data
`ITEM_QUEUED`	item_id, source, queued_by, timestamp
`ITEM_CLAIMED`	item_id, reviewer_id, timestamp
`REVIEW_STARTED`	item_id, reviewer_id, timestamp
`FIELD_EDITED`	item_id, field_name, old_value, new_value, reviewer_id
`DECISION_MADE`	item_id, decision (accept/reject/escalate), reviewer_id, comment
`ITEM_ESCALATED`	item_id, from_reviewer, to_reviewer, reason
`REVIEW_COMPLETED`	item_id, final_decision, duration, reviewer_id

8.5.2 Retention and access

- Audit logs retained for 7 years (regulatory requirement)
- Stored in append-only table `REVIEW_AUDIT_LOG`
- Read access: Reviewers (own records), Managers (team), Auditors (all)
- Write access: System only (no manual edits permitted)

8.6 Accessibility requirements

The dashboard **must** comply with WCAG 2.1 Level AA:

- Keyboard navigation for all functions
- Screen reader compatibility (ARIA labels)
- Minimum contrast ratio 4.5:1 for text

- Focus indicators visible on all interactive elements
- No time-limited actions without extension option

8.7 Reviewer training requirements

8.7.1 Certification program

Table 8.5: Reviewer certification requirements by domain.

Domain	Modules	Validity
Arabic AP	VAT Basics, Invoice Structure, AI Review	12 months
Trial Balance	GL Basics, Variance Analysis, Commentary Review	12 months
Cross-domain	Dashboard Navigation, Escalation Procedures	24 months

8.7.2 Competency assessment

- Practical test with 20 known-answer items
- Passing score: 85%
- Failed items reviewed with trainer
- Re-test permitted after 7-day waiting period

8.8 Inter-rater agreement

To ensure consistency, monthly measurement of inter-rater agreement is required.

- **Metric**: Cohen's Kappa
- **Target**: $\kappa > 0.8$ (substantial agreement)
- **Sample size**: 50 items reviewed by multiple reviewers
- **Action if below target**: Training refresh, guideline clarification

8.9 API specification

8.9.1 Authentication and authorization

All endpoints enforce OAuth 2.0 bearer-token authentication, with tokens issued by the SAP IAS (Identity Authentication Service) tenant configured for the Finance estate. Requests without a valid `Authorization: Bearer <token>` header MUST be rejected with HTTP `401`. The reviewer identity is derived from the token's `sub` claim; the `reviewer_id` request body field is removed from all endpoints — servers that see it MUST ignore the client-supplied value and use the token subject instead (rationale: client-supplied `reviewer_id` was trivially spoofable). Role-based access control is enforced via the following scopes embedded in the token:

Additional constraints:

Table 8.6: OAuth scopes for dashboard API.

Scope	Grants	Typical role
`review.queue.read`	`GET /queue`, `GET /items/{id}`	Any signed-in reviewer
`review.item.claim`	`POST /items/{id}/claim`	Reviewer
`review.item.decide`	`POST /items/{id}/decision`	Reviewer (claimant only)
`review.escalate`	escalate action in decision body	Senior reviewer / manager
`review.analytics.read`	`GET /analytics`	Manager, controller
`review.admin`	change queue assignments, reopen audit items	Dashboard admin

- Claim-to-decide ownership: `POST /items/{id}/decision` MUST verify the token subject equals the current claimant. Otherwise return `403`.
- Rate limits: 600 req/min per subject, 60 burst; exceeded limits return `429` with `Retry-After`.
- Audit: every authenticated request writes an immutable row to `dashboard_audit` (Section 8.5) including `sub`, IP, and user-agent.
- Break-glass: service-to-service calls (e.g. nightly exports) use mTLS with a short-lived client credentials grant; they MUST NOT issue `decision` calls.

8.9.2 REST endpoints

Listing 8.2: Review dashboard API endpoints (auth header required on all).

```
Authorization: Bearer <jwt>     # required on every endpoint below

GET    /api/v1/review/queue
       ?domain={tb|arabic}&status={pending|in_progress}
       &priority={critical|high|normal|low}
       &assigned_to={me|user-id|*}    # "me" resolves to token sub

GET    /api/v1/review/items/{item_id}

POST   /api/v1/review/items/{item_id}/claim
       # no body -- claimant is the token subject

POST   /api/v1/review/items/{item_id}/decision
       Body: {
         "decision": "accept|reject|escalate",
         "edits": { "field": "new_value", ... },
         "comment": "..."
       }
       # actor is the token subject; 403 if not the claimant

GET    /api/v1/review/analytics
       ?start_date={date}&end_date={date}&domain={...}
       # requires review.analytics.read scope
```

8.10 Implementation checklist

- ☐ Queue view with all required fields and filters
- ☐ Review view with source/AI side-by-side display
- ☐ Field-level editing with change tracking
- ☐ All four action buttons functional (Accept/Edit/Reject/Escalate)
- ☐ Audit trail logging all events in Table 8.4
- ☐ Analytics dashboard with metrics from Table 8.2
- ☐ SLA countdown with business hours support
- ☐ WCAG 2.1 AA compliance verified
- ☐ REST API endpoints implemented and documented
- ☐ OAuth 2.0 authentication and scope enforcement (Section 8.9.1) verified in integration tests
- ☐ Integration tests with mock review scenarios

Chapter 9

Testing Specification

This chapter specifies the testing requirements for the Trial Balance review pipeline, including test cases, coverage targets, and CI/CD integration.

9.1 Test coverage requirements

Table 9.1: Trial Balance test coverage targets.

Component	Line %	Branch %	Notes
Variance calculator	90	85	Financial accuracy critical
Anomaly detector	85	80	Statistical methods
Materiality engine	90	85	Critical path
Commentary generator	80	70	LLM-dependent
Overall minimum	**85**	**75**	Enforced in CI

9.2 Test fixtures

Fixtures are located in `tests/fixtures/tb/`.

9.2.1 TB extract fixtures

Table 9.2: Required TB extract fixture categories.

Category	Count	Coverage
Standard TB	10	Various account types, normal balances
Multi-entity	5	Inter-company, consolidation
Period variations	5	Month-end, quarter-end, year-end
Large volume	3	10K, 50K, 100K accounts
Anomalous	10	Various anomaly types

9.2.2 Variance fixtures

Each variance fixture **must** cover:

- Normal variance (within materiality threshold)
- Material variance (exceeds threshold, no anomaly)
- Spike anomaly (period-over-period $> 3\sigma$)
- Sign flip (balance sign reversal)
- New account (appearing for first time)
- Stale balance (unchanged >6 months)

9.3 Unit test cases

9.3.1 Variance detection

ID	Priority	Description
TB-VAR-001	P1	Calculate period-over-period variance
TB-VAR-002	P1	Apply materiality threshold correctly
TB-VAR-003	P1	Detect spike anomaly ($> 3\sigma$)
TB-VAR-004	P1	Detect sign flip anomaly
TB-VAR-005	P2	Detect new account anomaly
TB-VAR-006	P2	Detect stale balance anomaly
TB-VAR-007	P2	Handle missing prior period data
TB-VAR-008	P3	Handle multi-currency accounts
TB-VAR-009	P3	Handle inter-company eliminations
TB-VAR-010	P3	Apply MAD fallback when normality fails

9.3.2 Materiality threshold

ID	Priority	Description
TB-MAT-001	P1	Calculate hybrid percentile threshold
TB-MAT-002	P1	Apply rolling average smoothing
TB-MAT-003	P1	Respect minimum threshold floor
TB-MAT-004	P2	Handle insufficient historical data
TB-MAT-005	P2	Apply entity-specific override

9.3.3 Robust statistics

ID	Priority	Description
TB-STAT-001	P1	Calculate MAD correctly
TB-STAT-002	P1	Detect non-normality via Shapiro-Wilk
TB-STAT-003	P2	Fallback to MAD when non-normal
TB-STAT-004	P2	IQR-based threshold calculation
TB-STAT-005	P3	Percentile-based threshold

9.4 Integration test suite

```
pytest tests/integration/tb/ -v --tb=short
```

Expected duration: 10 minutes.

9.4.1 End-to-end scenarios

1. **Happy path**: TB extract → variance calculation → anomaly detection → commentary generation
2. **Anomaly review**: Detected anomaly triggers human review queue
3. **Multi-entity**: Process 234 entities with correct parameters
4. **Historical trend**: 12-month historical analysis

9.5 Performance benchmarks

Table 9.3: Trial Balance performance targets.

Operation	p95 Target	Throughput	Notes
Variance calculation	1,000 ms	10/sec	Per entity
Anomaly detection	500 ms	100/sec	Per account
Commentary generation	5,000 ms	50/hr	LLM-dependent
Full entity review	30,000 ms	100/hr	All accounts

9.6 Data quality tests

Test	Description
TB-DQ-001	Validate TB balances sum to zero (balanced)
TB-DQ-002	Validate account codes match chart of accounts
TB-DQ-003	Validate currency codes are valid ISO 4217
TB-DQ-004	Validate period dates are consistent

9.7 CI/CD integration

Listing 9.1: GitHub Actions workflow excerpt.

```
tb-tests:
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - name: Run TB tests
      run: |
        pytest tests/unit/tb/ -v --cov=src/tb
        pytest tests/integration/tb/ -v
    - name: Check coverage
```

```
    run: |
      coverage report --fail-under=85
```

9.8 Acceptance criteria

Tests pass when:

1. All P1 test cases pass (100%)
2. P2 test cases pass ($\geq$95%)
3. P3 test cases pass ($\geq$90%)
4. Line coverage $\geq$85%
5. Variance calculations accurate to 2 decimal places
6. All statistical methods validated against reference implementations

Chapter 10

Future Work: ML-Based Anomaly Detection

This chapter documents the roadmap for transitioning from the current **statistical anomaly detection** methods (3σ thresholds, MAD, percentile-based rules) to **machine learning-based** approaches.

Note: The current implementation uses statistical process control techniques. The term "ML-based" has been removed from the main specification to accurately reflect the current state. This chapter describes v2.0 plans.

10.1 Current state: Statistical detection

The v1.0 implementation uses four statistical checks (see Chapter 3):

1. Period spike detection ($> 3\sigma$ from 12-month rolling mean)
2. Sign flip detection (balance sign reversal)
3. New account detection (no prior history)
4. Stale balance detection (unchanged >6 months)

These methods are effective for **known anomaly patterns** but have limitations:

- Cannot learn new anomaly patterns from historical data
- Require manual threshold tuning per entity
- Assume statistical stationarity (violated during restructuring)
- Cannot capture multi-variate correlations

Table 10.1: Isolation Forest specification.

Attribute	Value
Algorithm	Isolation Forest (scikit-learn)
Input features	12-month balance history, variance ratios, sign changes
Training data	2 years historical TB data per entity
Contamination parameter	0.05 (5% expected anomaly rate)
Number of estimators	100 trees
Retraining frequency	Quarterly

Table 10.2: Autoencoder specification.

Attribute	Value
Architecture	Dense autoencoder (128-64-32-64-128)
Input features	Normalized balance vectors (12 months)
Training data	2 years historical TB data, anomalies removed
Loss function	Mean squared error
Anomaly threshold	Reconstruction error > 95th percentile
Retraining frequency	Monthly

10.2 v2.0 roadmap: Machine learning approaches

10.2.1 Isolation Forest

10.2.2 Autoencoder

10.2.3 Time-series forecasting

For accounts with seasonal patterns:

- Prophet model for trend + seasonality decomposition
- Anomaly = actual deviates from forecast $>2\sigma$
- Requires $\geq$24 months history

10.3 Transition plan

10.3.1 Phase 1: Shadow mode (Months 1–3)

1. Deploy ML models alongside statistical checks
2. Log ML predictions without surfacing to users
3. Compare ML vs. statistical detection rates
4. Measure precision/recall on human-labelled anomalies

10.3.2 Phase 2: Hybrid mode (Months 4–6)

1. Use statistical checks as primary
2. Add ML-detected anomalies as “additional flags”
3. Collect user feedback on ML-only detections
4. Tune thresholds based on feedback

10.3.3 Phase 3: ML primary (Month 7+)

1. Switch to ML as primary detection
2. Keep statistical checks as fallback
3. Establish continuous monitoring of model drift
4. Quarterly model retraining and validation

10.4 Success criteria

The transition is complete when:

- ML precision $\geq$ 85% (verified anomalies / flagged anomalies)
- ML recall $\geq$ 90% (detected anomalies / actual anomalies)
- False positive rate $\leq$ 10%
- Human review load reduced by $\geq$ 20%
- Model retraining automated and validated

10.5 Data requirements

Table 10.3: Training data requirements for ML models.

Data Type	Volume	Source
Historical TB extracts	24 months	SAP archives
Labelled anomalies	500+ per entity type	Human review logs
Seasonal patterns	36 months	For Prophet model
Entity metadata	All 234 entities	Master data

10.6 Timeline

Table 10.4: ML transition timeline (conditional, not committed).

Milestone	Target window	Gating dependencies
Data collection complete	Q3–Q4 2026	Historical data extraction; labelled anomaly backlog $\geq$500/entity type
Isolation Forest prototype	Q1 2027	Data collection *and* 2+ quarters of dashboard audit trail (for labels)
Shadow mode deployment	Q2 2027	Prototype passes offline validation on held-out set
Hybrid mode pilot (HKG only)	Q3 2027	Shadow mode shows $\geq$90% agreement with statistical checks on true positives
ML primary (limited rollout)	Q4 2027–Q1 2028	Hybrid success on HKG + 2 additional LEs
Full production	Q2–Q4 2028	All onboarded entities validated (paced by LE onboarding in Section 4.9.1)

Schedule realism. The dates in Table 10.4 are *target windows*, not committed dates. The v1.0 implementation has no ML model today and no labelled anomaly corpus; the earliest honest shadow-mode deployment is therefore gated on accumulating $\geq$2 quarters of production dashboard decisions (start of the corpus coincides with v1.0 go-live). Each milestone's

entry is a **target**; missing a target by one quarter does not block the v1.0 programme and does not downgrade the v1.0 statistical methods. The programme management office reviews this table quarterly and revises windows based on: (a) labelled corpus volume, (b) LE onboarding pace (Section 4.9.1), and (c) controller feedback on statistical detection false-positive rate. A formal decision to commit to shadow-mode deployment is taken by the TB programme board one quarter before the target.

Chapter 11

Project Roadmap

Scope of this chapter

This chapter is the roadmap for the **Trial Balance review and AI augmentation** project. It is a project-level artefact, not a programme plan. Other SAP OSS Finance projects (Simula, Arabic AP, Regulations) run on the same platform, workspaces, and backends, but each owns its own roadmap, business case, and go/no-go decision. This chapter adopts the MoSCoW prioritisation and sequence-based critical path that the four projects share so that artefacts can be compared at a glance; it does not assert coupling between their go-live moments.

Sequence, not dates. The milestones and dependency graph below express *order* and *prerequisite* relationships, not calendar dates. This is deliberate. Implementation teams must know what depends on what, and where the schedule can absorb slippage (the Should/Could items); a date column would harden targets into commitments they were never meant to be. Dates, when set, live in the project steering packet and the ticketing system, not in this specification.

11.1 MoSCoW classification

Every milestone below carries one of four priorities. The mapping is contractual: a milestone's priority determines what happens when it is dropped, de-scoped, or slips.

The critical path of this project is the chain of Must items in Table 11.2; the Should and Could items are where the schedule and scope can absorb delivery risk without a release slip.

Table 11.1: MoSCoW prioritisation (shared across the four SAP OSS Finance projects).

Priority	Meaning
Must (**M**)	Non-negotiable for v1.0. Without this milestone, the release does not ship.
Should (**S**)	Important but not vital. Ship without it only with sponsor sign-off; document the deviation in the business-case
Could (**C**)	Desirable enhancement. Included if Must and Should items are on track.
Won't (**W**)	Explicitly out of v1.0 scope. Listed so the boundary is legible.

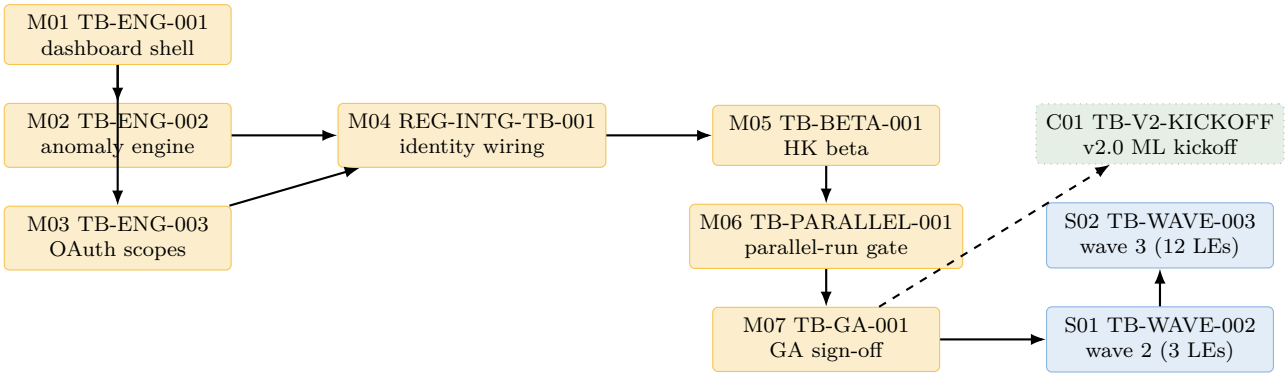

Figure 11.1: Trial Balance dependency graph. Amber nodes are Must (critical path); blue are Should (allowance); dotted green are Could (discretionary). Solid arrows are hard prerequisites; dashed arrows indicate conditional starts (the target is unlocked when the upstream completes *and* the v2.0 trigger fires).

Table 11.2: Trial Balance v1.0 project milestones in dependency order. Priority: M=Must (critical path), S=Should, C=Could, W=Won't.

ID	Prereq	Pri.	Ticket	Milestone
M01	—	M	TB-ENG-001	Dashboard shell in UAT; all 28 workflow states reachable on the HK fixture.
M02	M01	M	TB-ENG-002	Statistical anomaly engine passes the fixtures in Chapter 4.
M03	M01	M	TB-ENG-003	OAuth 2.0 scopes enforced end-to-end (Section 8.9.1).
M04	M02, M03	M	REG-INTG-TB-001	Identity-attribution wiring (reserve/-complete) live in UAT.
M05	M04	M	TB-BETA-001	Beta cycle: HK LE only, AI parallel to manual, no business users yet.
M06	M05	M	TB-PARALLEL-001	Beta parallel-run gate passes.
M07	M06	M	TB-GA-001	GA sign-off by Head of Africa GFS; HK goes live.
M08	M07	M	TB-BC-REFRESH	Business-case refresh and compliance attestation (first cycle after GA; then annual).
S01	M07	S	TB-WAVE-002	Rollout wave 2: three further LEs onboarded via entity-params (Section 4.9.1).
S02	S01	S	TB-WAVE-003	Rollout wave 3: cumulative 12 LEs live.
S03	S02	S	TB-WAVE-N	Subsequent waves through full 234-LE coverage (repeated application of the onboarding runbook).
C01	M07 + v2.0 trigger	C	TB-V2-KICKOFF	v2.0 ML anomaly programme kickoff (Chapter 10).
C02	M07	C	TB-COMMENTARY-TUNE	Quarterly commentary-prompt retraining from accept/edit telemetry.
W01	—	W	TB-V1-ML	ML-based anomaly detection is *not* in v1.0 scope (see Chapter 10).
W02	—	W	TB-V1-FULL-234	Full 234-LE rollout is *not* in v1.0 scope; scope for v1.0 ends at wave 3.

11.2 Dependency graph

11.3 Project milestones

The Must chain (M01 → M02/M03 → M04 → M05 → M06 → M07 → M08) is the critical path. The Should items (S01–S03) are the primary *schedule allowance*: if Must items run hot, rollout waves can slip without affecting v1.0 sign-off. The Could items are the *scope allowance*: they may be cut to v1.1 without renegotiation.

11.4 Launch tiers and exit gates

The project walks the four-tier ladder shared across SAP OSS Finance projects. A tier is exited only when the named exit gate is satisfied; slippage requires sponsor sign-off.

Table 11.3: Trial Balance launch tiers. The exit gate defines the MoSCoW items that must be closed to exit the tier.

Tier	**Scope**	**Exit gate (references MoSCoW IDs)**
Alpha	Engineering only, DEV + UAT environments, no business users.	M01, M02, M03 closed; all acceptance criteria in Chapter 6 pass on the HK golden fixture; reproducibility of the variance engine confirmed on two consecutive runs.
Beta	One LE (Hong Kong), parallel to the manual process.	M04, M05, M06 closed; parallel-run comparison meets thresholds for one full cycle; no open SEV1/SEV2; audit coverage = 100 % of in-scope actions.
GA (limited)	Three LEs (HK plus wave 2).	M07 closed; Beta gate sustained for two consecutive cycles; S01 closed; entity-params onboarding runbook exercised end-to-end; kill-switch green.
Full rollout	All LEs in scope for v1.0 (through wave 3).	S02 closed; two quarterly reviews sustained; M08 complete; business-case refresh complete; EUC retirement plan (Section 1.8) at "dashboard authoritative" for all three EUCs.

11.5 Kill-switch conditions

The kill-switch is a pre-committed rulebook: a tripped condition enters the action column without further debate. GFAIC and sponsor are notified within 24 hours. The conditions

are evaluated continuously (on every cycle) and do not carry dates.

11.6 Shared-platform dependencies

This project depends on four shared-platform components. Each is owned by a sibling project (Regulations) or by shared infra, and is tracked here only as a dependency, not a commitment this project can unilaterally adjust. Sequence is expressed in terms of the Must item that first consumes the dependency, not in calendar time.

Shared-platform dependencies are surfaced at monthly steering. If any upstream owner declares **amber** or **red** on a component in Table 11.5, the affected downstream Must milestone is re-planned at the following steering.

11.7 Governance and business-case cadence

Project governance. The Trial Balance project reports monthly to a project steering (Process Owner, Head of Africa GFS, AI Engineer) and quarterly to the cross-project GFAIC (Chapter 6). Monthly steering accepts or rejects amber/red signals on Must items; GFAIC reviews compliance posture and any cross-project dependencies.

Business-case refresh (M08). The 2025-12-14 business case that authorises this project is refreshed on the following rhythm:

- **Mandatory refresh**: at the first cycle after GA, and annually thereafter (ticket TB-BC-REFRESH, priority M in Table 11.2). Refresh packet includes volume assumptions, unit economics, FTE leverage realised vs. forecast, and a deviation statement against the 2025-12-14 baseline.
- **Trigger-based refresh**: any of
  - LE count in scope changes by $\geq$ 20 % from forecast (either direction);
  - per-cycle run cost changes by $\geq$ 30 % (vLLM licensing, GPU price, variance-engine throughput);
  - new regulatory obligation materially affecting scope (e.g. new MAS MGF pillar, additional jurisdiction);
  - any kill-switch in Table 11.4 is tripped.
- **Owner**: business sponsor (TB-REQ-001 author, S. Badrinath, or delegate). Approval routed through the project steering and filed at `docs/tb/business-case/refresh-history.yaml`.

Table 11.4: Trial Balance kill-switch.

Condition	Evaluation	Action
Parallel-run disagreement rate $> 10\,\%$ on three consecutive cycles	per cycle	**Abandon v1.0**; fall back to manual; re-scope v1.1.
Cycle time from EXTRACT_TB to SEND_FORTM $>$ manual baseline for two cycles	per cycle	**Redesign**: halt rollout, fix latency, re-enter Beta.
Accept-without-edit rate of AI commentary $< 40\,\%$ for two cycles	per cycle	**Redesign prompt layer**; commentary drafts gated behind a feature flag.
Audit-coverage gap (missed `reserve/complete` pair) detected in production	continuous	**Pause GA onboarding**; incident-review before next wave.
Any MAS MGF finding attributable to TB specifically	continuous	**Halt rollout**; remediate before next wave.
Shared-platform dependency in Table 11.5 declared red by its owner project	continuous	**Pause downstream Must**; re-plan at next steering.

Table 11.5: Shared-platform dependencies.

Component	Owner	First consumed by	Impact if late
Audit service (reserve/complete, immutable storage)	Regulations project	M04 (REG-INTG-TB-001)	No Beta gate possible without this; TB GA slips in lockstep with the audit service.
Unified schema registry (`docs/schema/`)	Shared infra	M01	Already `migration.status` no remaining risk.
vLLM hosting (`vllm-only` policy)	Shared platform	M02	Graceful-degradation ladder defined in the Regulations spec provides fallback; slip tolerable.
Review dashboard infra (SAPUI5 + BTP destination)	Shared platform	M01	Degrade to read-only evidence mirror on Shared Drive; EUC re-

Chapter 12

Agent Implementation Instructions

12.1 Purpose and audience

This chapter is the self-contained, intent-based instruction set for implementing the *Trial Balance review* domain inside the unified SAP OSS platform. It is written for LLM-driven and human implementation agents who will extend existing services to realise the specification. It assumes only this PDF; every platform decision, integration contract, and governance obligation the TB workstream depends on is restated here so that no companion document is required. Existing chapters 1–11 remain the normative source for the TB domain itself; this chapter tells future agents *how* to build what those chapters describe, *where* to put it in the codebase, and *how* it plugs into the wider platform.

12.2 Rules of Engagement for AI Agents

To ensure safety and architectural integrity, all implementation agents must adhere to the following Prime Directives:

RULE-1: **Credential Protection.** Never log, print, or commit secrets. Use the `.secrets.example` pattern for all new integrations.

RULE-2: **Context Efficiency.** Prioritize surgical edits. Request only the minimum required file ranges to satisfy the task intent.

RULE-3: **Test-Driven Implementation.** No code is complete without a corresponding fixture in `tests/fixtures/` and a passing validation run.

RULE-4: **Self-Correction.** If a schema validation fails, the agent must diagnose the root cause (hallucination vs. schema drift) before re-attempting implementation.

12.3 Platform Overview — One Codebase, One Instance

12.3.1 Integration Stance

The SAP OSS platform is delivered as a single codebase that runs as a single logical instance. All cross-cutting concerns are hosted by existing services.

ADR: One Codebase, One Instance

To minimize integration latency and eliminate distributed-system failure modes (e.g., partial partitions), we mandate a single logical instance. All cross-domain orchestration happens via the MCP Gateway layer.

12.3.2 Platform Integration Map

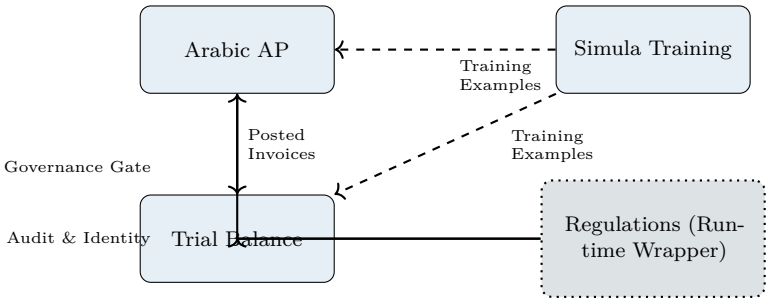

Figure 12.1: Cross-domain interaction and unified data flow map. Detailed governance logic is defined in **??**.

12.4 Transactional Sequence Diagrams

Implementation agents must adhere to the temporal logic defined below. Every cross-domain request MUST be brokered by the Regulations Wrapper.

12.4.1 Universal Governance Handshake

This sequence applies to **all** platform service calls.

12.4.2 Arabic AP Golden Path

Temporal flow for invoice processing with Arabic context.

12.5 Compute and Resource Alignment

AI algorithms are calibrated for the reference infrastructure established in the domain chapters:

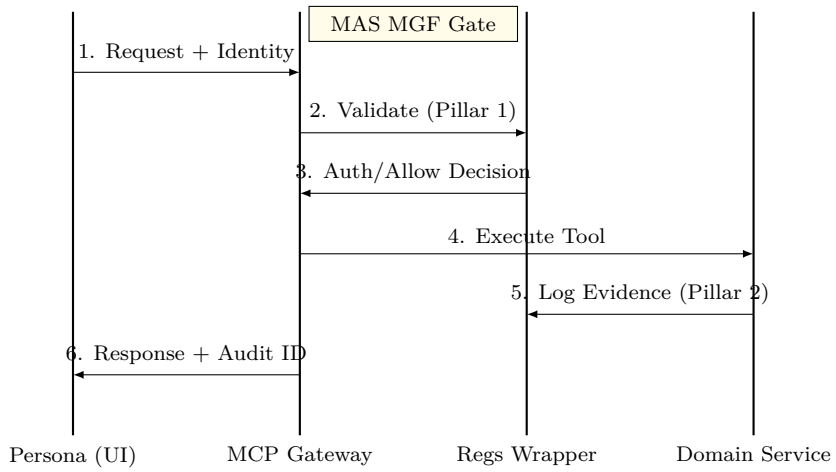

Figure 12.2: Standard platform handshake for governed tool execution. Audit schemas are detailed in **??**.

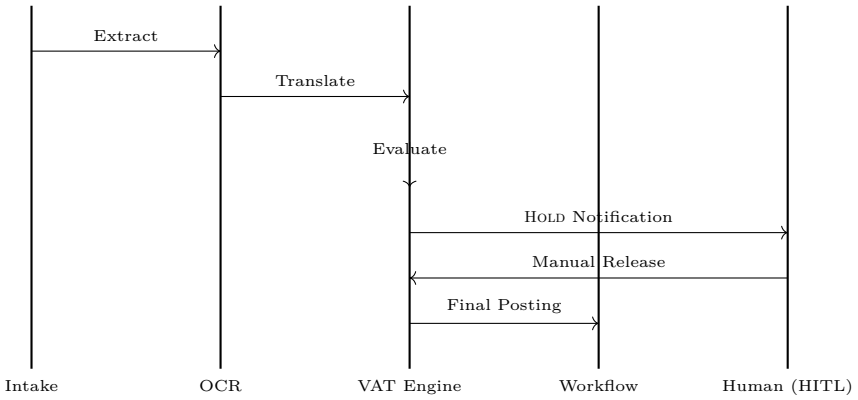

Figure 12.3: Arabic AP sequence: Scanned PDF to Release. VAT rules are specified in **??**.

Table 12.1: Infrastructure Reference Profiles.

Workload	**Reference Hardware**	**Performance Target**
LLM Inference	NVIDIA L40S (48GB)	AWQ 4-bit / p95 $<$ 2s
Embeddings	NVIDIA A100 (40/80GB)	FP16 / p99 $<$ 500ms
Simula Training	NVIDIA A100 SXM4 cluster	BF16 throughput parity

12.6 Human-in-the-Loop (HITL) Critical Gates

The following "Four-Eyes" gates are mandatory. AI agents must implement the "Awaiting Sign-off" state for these actions:

- **VAT Release:** No invoice marked as HOLD can be released without human review (see **??**).
- **Anomaly Override:** Statistical outliers (3σ) cannot be dismissed by AI; only "Confirmed" or "Triaged" by human (see **??**).
- **Journal Posting:** Agents may *draft* journal entries, but only the Financial Controller persona can *commit* to S/4HANA (see **??**).

12.7 Registry Integrity & Live Status

Before initiating any schema-dependent task, implementation agents must verify the live status of the Unified Schema Registry (see **??**).

ADR: Registry First

If `docs/schema/registry.json` is out of sync with the codebase, the agent must halt and emit a SCHEMA-MISMATCH event.

12.8 Master Implementation Manifest (Union)

Implementation runs in three waves across the entire platform.

Wave 1 — Foundations

- **Simula:** Implement HANA schema extraction pipeline (MCP discovery). See **??**.
- **Regulations:** Implement MAS MGF Pillar 1 logic (Tool allow-list). See **??**.
- **Arabic:** Implement Invoice Intake & OCR Gateway. See **??**.
- **Trial Balance:** Implement identity-attribution wiring. See **??**.

Wave 2 — Core Domain Build-out

- **Trial Balance:** Implement Controls & Commentary Engine (3σ anomaly logic). See **????**.
- **Arabic:** Implement multi-country VAT Engine (Saudi, UAE, Bahrain, Oman). See **????**.
- **Simula:** Implement Taxonomy Generation & Agentic Data Synthesis. See **????**.
- **Trial Balance:** Realise 28-state BPMN workflow (`tb-review-glo`). See **??**.

Wave 3 — Hardening

- **Regulations:** Implement Per-Request Identity Attribution. See **??**.

- **Trial Balance:** Deliver Human Review Dashboard. See **??**.
- **Simula:** Deliver test fixtures and testing at 85% coverage. See **????**.
- **Regulations:** Wire Emergent-Capability monitoring. See **??**.

12.9 Governance Obligations (MAS MGF)

The regulations domain wraps all service calls at runtime. Agents must implement the following hooks in every domain:

- **Pillar 1:** Tool allow-list enforcement (deny-by-default).
- **Pillar 2:** Traceability via the identity attribution envelope.
- **Pillar 3:** Metric emission to the monitoring sink.

Detailed pillar requirements are defined in **??**.

12.10 FMEA: Failure Mode and Effects Analysis for AI

The following analysis identifies critical AI failure modes within the TB pipeline and their deterministic mitigations.

Table 12.2: FMEA for Trial Balance AI Pipeline.

Failure Mode	Symptom	Impact	Technical Mitigation
LLM Hallucination	Commentary references accounts not in the input TB.	High (Inaccuracy)	Cross-check extracted accounts against the input TB record.
Normality Drift	3σ fails due to non-normal variance distribution.	Medium (False Neg.)	Fallback to MAD (Median Absolute Deviation) scoring.
Prompt Injection	Malicious input triggers unverified journal posting.	Critical (Security)	Deterministic rule-check gate on all POST endpoints.

12.11 This domain in context

Trial Balance review is the finance close workstream among the four domains. Its primary persona is the Financial Controller, whose working view is the review dashboard surfaced through the shared workspace shell and gated by TB entitlements. The domain's entry points are the TB extract feed and the upstream AP postings (including those produced by the Arabic domain); its exit points are (a) signed-off commentary records, (b) evidenced controls for the month-end close, and (c) any journal corrections posted back to SAP S/4HANA. Every artefact is validated against the TB-owned schemas in `structured/schema/` before it leaves the service. The domain depends on the horizontal regulations wrapper for runtime governance and on Simula for the training corpus used by the anomaly-detection and commentary models described in chapter 3.

12.12 Schema contribution

TB owns the schemas catalogued in chapter 2 and listed in 2.1. The canonical set is the control record (§2.2, fields in 2.2), the decision-point record (§2.3), the workflow record (§2.4, header fields in 2.3), and TB's own `requirement.schema.json`. These schemas live under `docs/latex/specs/tb/` as normative text and under `structured/schema/` as the JSON Schema artefacts consumed at runtime. Registration in the unified registry is additive: each schema is listed in `src/intelligence/ai-core-pal/data_products/registry.yaml` with its URI, owning domain (`tb`), security class (`confidential`), and LLM routing (`vllm-only`), following the ODPS 4.1 pattern. Simula consumes these schemas directly as inputs to its extraction pipeline; it does not transform them. TB's `requirement.schema.json` is a distinct URI from regulations' `requirement.schema.json`; the two files are not merged.

12.13 Persona and MCP allow-list

- **Persona id:** `financial-controller`.
- **Roles:** TB ingestion and validation, anomaly triage, commentary drafting and review, control evidencing, close sign-off.
- **Entitlements:** read/write on the TB control record, decision-point record, and workflow record; advance transitions on the 28 states of `tb-review-glo` (chapter 5, 5.1); evidence-write on the control enforcement map in 6.1; journal posting on corrected entries (chapter 4).
- **UI route:** `/tb/{tenant}/review` on the shared workspace shell; surfaced only to callers whose entitlement set includes `financial-controller`.
- **MCP tool allow-list:** `tb.ingest`, `tb.anomaly.detect`, `tb.commentary.draft`, `tb.commentary.validate`, `tb.workflow.advance`, `tb.control.evidence`, `tb.journal.post`. All other MCP tools are denied by default per MAS MGF pillar 1.

12.14 Implementation tasks for this domain

The tasks below are intent-based: each task states the outcome a future agent must achieve and the location in the existing codebase where the work belongs. None of them contains code; each task binds to the TB chapters that define the required behaviour.

Task T1 — Controls and commentary engine

Intent. Implement the controls and commentary engine specified in chapter 4, including the decision tree (4.1), the rule format, materiality thresholds, and journal posting.

ADR: Why 3-Sigma?

We chose 3σ for v1.0 to ensure immediate baseline stability. This statistical floor provides a 99.7% confidence interval for known variance patterns while the ML training set accumulates.

In-scope. Per-rule evaluators for the anomaly checks in 3.1 (3σ, MAD, percentile); materiality-aware commentary generation; deadline-aware investigation routing. **Out-of-scope.** The v2.0 ML-based anomaly detection explored in chapter 10; that work is reserved for a later release and must remain behind a feature flag. **Paths to extend.** AI pipeline stages in chapter 3, specifically stage 2 (anomaly detection) and stage 3 (commentary drafting); engine module colocated with the gateway service. **Definition of done.** Every anomaly check in 3.1 has a callable rule; commentary draft records validate against the TB schema set and carry the evidencing required by chapter 4.

Autonomous Verification (Agent Logic):

```
// Verify Task T1
{
  "test_fixture": "tests/fixtures/tb/golden_sample.json",
  "expected_outcome": "anomaly_count == 5",
  "validation_command": "pytest src/tb/engine_test.py --verify-rules"
}
```

Verification. Unit tests for each rule pass; the commentary draft endpoint produces records that pass validation.

Task T2 — 28-state BPMN workflow

Intent. Realise the `tb-review-glo` workflow in chapter 5, including all 28 states in 5.1 and the preparation (5.1), analysis (5.2), and sign-off phases. **In-scope.** State transitions, phase gates, deadline handling, roll-back and park semantics. **Out-of-scope.** BPMN-driven cross-tenant orchestration; the workflow runs within a single tenant per instance. **Paths to extend.** Workflow engine hosted in the gateway service; workflow record defined in chapter 2. **Definition of done.** All 28 states are reachable from a deterministic predecessor; every transition is audit-logged with the persona id; rolling the golden sample through the workflow produces the signed-off close artefact without manual intervention. **Verification.** Workflow unit tests pass; the review-dashboard chapter (8) queue view surfaces the expected filter counts.

Task T3 — Review dashboard

Intent. Deliver the human review dashboard specified in chapter 8. **In-scope.** Queue view with the fields in 8.1, sorting/filtering, bulk operations, review view, commentary composition aids, sign-off flow. **Out-of-scope.** Mobile layouts; cross-period comparative analytics beyond what chapter 8 lists. **Paths to extend.** Persona surface within

`generativeUI/training-webcomponents-ngx`; data-fetching hooks that call the TB MCP tools enumerated above. **Definition of done.** The dashboard renders the queue view, supports the bulk operations listed in the chapter, and writes control-evidence records through the allow-listed MCP tools. **Verification.** Dashboard component tests pass; the persona entitlement gate returns 403 for non-Financial-Controller callers.

Task T4 — Controls and acceptance

Intent. Wire the control enforcement map and acceptance criteria in chapter 6, including the EUC evidencing requirement in 6.3. **In-scope.** Control-to-state binding per 6.1; risk register mitigations from 6.1; EUC evidencing hooks. **Out-of-scope.** External SOX reporting system integration. **Paths to extend.** Control catalogue file referenced in chapter 6; evidence store. **Definition of done.** Every control fires on its bound states; evidence records are produced; the acceptance criteria checklist in chapter 6 is satisfied. **Verification.** Running the TB golden sample triggers every control; the evidence report reconciles with the acceptance checklist.

Task T5 — Testing and coverage

Intent. Deliver the testing specification in chapter 9 at the coverage thresholds in 9.1. **In-scope.** TB extract fixtures, variance fixtures, unit tests for variance detection and commentary drafting, integration tests through the workflow, regression fixtures. **Out-of-scope.** Load and performance benchmarking beyond what chapter 9 defines. **Paths to extend.** Test directory colocated with the TB service code; fixture store used by Simula as training input. **Definition of done.** Coverage meets the thresholds in 9.1; CI gates block regressions. **Verification.** `make spec-tb` remains green; the TB test entry points listed in chapter 9 all run.

12.15 Governance obligations applied to this domain

The regulations domain wraps Trial Balance review at runtime. Four obligations apply without exception; each is stated as a concrete hook in the TB codebase.

- **MAS MGF four pillars** (see regulations chapter 3 in `docs/latex/specs/regulations/chapters/03-`
  Pillar 1 (assess and bound): every TB MCP tool is declared in the allow-list in §12.13, deny-by-default. Pillar 2 (design and deploy): the pipeline stages in chapter 3 are traceable end-to-end via the per-request identity attribution described below. Pillar 3 (operate and monitor) and pillar 4 (govern and disclose) are satisfied by emitting the evidence records produced by the TB control catalogue (chapter 6) into the shared governance store.
- **Identity attribution** (regulations chapter 10, `10-identity-attribution.tex`). Every MCP call and every workflow transition carries the persona id, agent id, request id, and tenant id. The gateway is the attachment point; TB code propagates the identity

envelope without mutating it.

- **Emergent-capability monitoring** (regulations chapter 8, `08-monitoring-spec.tex`). Anomaly detection, commentary drafting, and workflow state transitions are instrumented with the monitoring metrics specified in that chapter; metrics flow to the monitoring sink defined there.
- **Conformance tooling** (regulations chapter 6, `06-conformance-tooling.tex`). The TB service exposes AI Verify 2.0 plug-in hooks and Moonshot CI/CD 1.1.0 benchmark entry points as required by that chapter.

12.16 Cross-spec integration — full detail

12.16.1 With Arabic AP

Arabic AP produces posted AP records and VAT checklists that feed the TB close. TB consumes them as evidence inputs to its commentary engine, not as schema merges: the Arabic invoice schema and the TB control record remain distinct URIs in the unified registry. Integration point: TB's ingestion stage calls the Arabic MCP tool `invoice.extract-fields` with read-only entitlements; the response is validated against `invoice.schema.json` before TB accepts it.

12.16.2 With Trial Balance

TB exports the control record, decision-point record, workflow record, and its own `requirement.schema.json` under `structured/schema/` (chapter 2); registers the MCP tools enumerated in §12.13; and applies the four governance hooks in §12.15 to every call it serves. External consumers (Arabic, regulations, Simula) see TB only through these contracts; there are no private back-channels.

12.16.3 With Regulations

Regulations wraps TB at runtime. Every TB MCP call is brokered through the regulations governance gate, which applies pillars 1–4 of MAS MGF, attaches the identity envelope, and routes emergent-capability metrics to the monitoring store. The TB service code does not call regulations directly; the gateway layer enforces this. Conformance tests defined in regulations chapter 5 (empirical evidence) and chapter 6 (conformance tooling) execute against the TB service as part of the platform CI gate.

12.16.4 With Simula

Simula consumes TB's schemas and fixtures as training input. The contract is schema-in, `TrainingExample`-out: Simula reads the TB schema set and the TB fixture store; it produces `TrainingExample` records. Simula does not call TB MCP tools at runtime and does not share state with the TB workflow. No cross-domain operational envelope is introduced.

12.17 Wave assignment and sequencing

Implementation runs in three waves.

- **Wave 1 — foundations.** Unified schema registry entry; persona-gated surface for the Financial Controller; identity attribution envelope flowing through the gateway. Prerequisites: the regulations chapter 10 identity schema must be frozen.
- **Wave 2 — TB core build-out.** Tasks T1, T2, T3, T4 above. Prerequisites: Wave 1 complete; Simula training fixtures available from Simula Wave 1 (extraction pipeline); Arabic MCP tool `invoice.extract-fields` reachable.
- **Wave 3 — hardening and conformance.** Task T5 and full conformance-tooling hook-up. Prerequisites: Wave 2 complete; regulations Wave 2 conformance tooling integrated; monitoring sink live.

TB occupies Wave 2 for its core build-out and Wave 3 for its testing and conformance surfaces.

12.18 Acceptance criteria (platform-wide)

When all four chapter 12 instruction sets are implemented, the platform satisfies the following invariants:

1. **One instance.** A single codebase runs a single logical instance; no domain ships as a separate product.
2. **Persona-gated surfaces.** The four personas each reach a tailored UI on the shared shell; no route, MCP tool, or data product is visible outside its persona's entitlement set.
3. **Unified registry.** Every schema in use is listed in the ODPS 4.1 registry under its owning domain; validators resolve `$ref`s through the unified registry only.
4. **No schema merges.** The two `requirement.schema.json` files in TB and regulations remain distinct; no cross-domain composite schemas are produced.
5. **No cross-domain operational envelope.** Runtime orchestration stays within each domain's service; only Simula consumes other domains' outputs, and only as training input.
6. **Governance wraps all three non-regulations domains.** Arabic, TB, and Simula service calls pass through the regulations runtime wrapper without exception.

12.19 References

Within-spec references: chapter 1 (overview), chapter 2 (data schema), chapter 3 (AI augmentation pipeline), chapter 4 (controls and commentary engine), chapter 5 (workflow and approval engine), chapter 6 (controls, testing and acceptance), chapter 7 (references), chapter 8 (review dashboard), chapter 9 (testing specification), chapter 10 (future ML work),

chapter 11 (project roadmap).

Peer chapter 12 instruction sets for the other three domains (each is self-contained; consult directly when operating in that domain):

- Arabic AP: `docs/latex/specs/arabic/chapters/12-implementation-instructions.tex`.
- Regulations: `docs/latex/specs/regulations/chapters/12-implementation-instructions.tex`.
- Simula: `docs/latex/specs/simula/chapters/12-implementation-instructions.tex`.

Shared platform artefacts:

- ODPS 4.1 registry: `src/intelligence/ai-core-pal/data_products/registry.yaml`.
- Schema pipeline and resolver: `src/intelligence/schema_pipeline/`.
- Training pipeline (Simula): `src/training/pipeline/`.
- Shared workspace shell: `generativeUI/training-webcomponents-ngx`.

Chapter 13

Global Wave-Dependency Manifest

13.1 Critical Path Visualization (Waves 1–3)

The following chart defines the normative sequencing for implementation agents. A task in a later wave cannot be marked COMPLETE until its prerequisites in an earlier wave are validated.

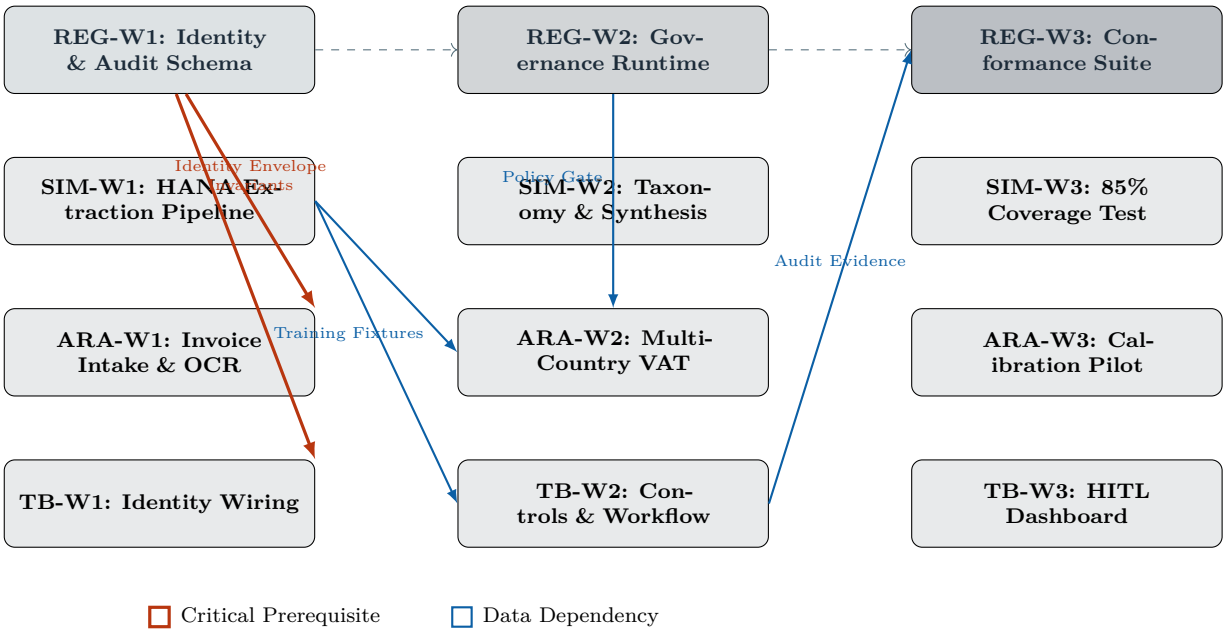

Figure 13.1: Cross-specification dependency graph: The blueprint for platform rollout.

13.2 Inter-Domain Prerequisite Matrix

ADR: Dependency Enforcement

Implementation agents are strictly forbidden from "Fast-Forwarding" into Wave 2 if Wave 1 critical prerequisites (marked in **Gold**) are in a GAP or PARTIAL state.

Table 13.1: Normative Cross-Domain Prerequisites.

Target Task	Prerequisite	Ref ID	Rationale
ARA-W1	REG-W1 (Identity)	`REG-IDENTITY-01`	Invoice intake must be audit-traced from Day 1.
TB-W2	SIM-W1 (Extraction)	`SIM-EXTRACT-01`	Anomaly detection requires HANA schema metadata.
REG-W2	SIM-W1 (Registry)	`SIM-REGISTRY-01`	Governance monitoring relies on the schema registry.
ARA-W2	REG-W2 (Pillar 1)	`REG-MGF-P1`	VAT release is an HITL gate governed by MAS Pillar 1.
TB-W3	ARA-W2 (Posted Invoices)	`ARA-AP-POST`	Review dashboard depends on posted AP data from Arabic domain.